

Parenting Decisions and Child Skill Development

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Abstract

This paper develops and estimates a dynamic model of child cognitive and non-cognitive skill development. In this framework, both parents and their children make decisions that influence skill development, with the novel feature that the children's preferences are endogenous and a function of parenting style choices made by their parents. Counterfactual analysis shows that if a policy can be designed to exogenously increase child study time, parenting style decisions change, cognitive skills are increased, but non-cognitive skills remain mostly unchanged.

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JEL Classification: J13, D1

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1 Introduction

The relationship between parents and their children is fundamental for a child’s growth and development. This importance advocates for a sound understanding of a number of different investment decisions parents make when raising their children. Since [Becker and Tomes \(1979\)](#), economists have placed increasing attention on the skill development of children, however, until recently, models have ignored the role children play. While economists have made great progress building upon work laid out by [Todd and Wolpin \(2003\)](#), [Cunha and Heckman \(2007\)](#), and [Cunha et al. \(2010\)](#) to understand how parental time and monetary investments influence child development, omitting children from the framework means that we ignore an important investment decision that parents make, the style in which they choose to parent. This omission opens up avenues for economists to explore not only how to introduce parenting style into an economic model, but to understand its role in shaping children’s cognitive and non-cognitive skills.

In this paper, I develop and estimate a dynamic model of children’s cognitive and non-cognitive skill development by introducing parenting style into a model of parental time allocation in the style of [Del Boca et al. \(2014\)](#). Similar to their more recent work in [Del Boca et al. \(2026\)](#), both parents and their child are active participants in the skill development process, however, our definitions of parenting style are distinct. This paper presents a novel way to use data from the Child Development Supplement (CDS) of the Panel Study of Income Dynamics (PSID) to classify the three major parenting styles outlined by [Baumrind \(1966\)](#), authoritarian, authoritative, and permissive. Parenting style decisions are allowed to influence the non-cognitive skills of children, which are linked to their preferences, making them endogenous¹. This allows for dynamics spillovers, as parenting style decisions can have lasting effects that will influence both child and parental behavior.

These three parenting styles are distinct and care must be taken when thinking about how to build them into an economic model. A permissive parenting style is characterized by allowing children to make their own choices with minimal parental interference. Authoritarian and authoritative parenting styles are more involved in the sense that parents set limits and rules for their children to follow, but, they differ in the ways these rules are monitored and enforced ([Baumrind \(1966\)](#) and [Baumrind \(1971\)](#)). An authoritarian parenting style is characterized by parental demand for obedience and control, and corporal punishment is often used in the event that rules are broken. An authoritative parenting style differs from a permissive parenting style in that rules are still set and enforced, but parents give their children an explanation for why the rules are in place instead of demanding obedience, and

¹See [Becker \(1996\)](#) for a detailed discussion about how economists can account for changing tastes.

do not utilize harsh punishment in the event that rules are broken, making it distinct from an authoritarian parenting style.

[Doepke and Zilibotti \(2017\)](#) present a theoretical model of these three parenting styles. They describe authoritarian and authoritative parenting styles as “intensive”, meaning that the parents exert some effort to either constrain their child’s choice set (authoritarian), or manipulate their preferences (authoritative). To capture the idea that rules are set and enforced when parents adopt an authoritarian or authoritative parenting style, I allow both of these parenting styles to constrain the choices of children, while children are free to make their own time allocation decisions when they face a permissive parenting style. In this paper, I also allow each parenting styles to influence the non-cognitive skill development process instead of only reserving preference manipulation for an authoritative parenting style.

There has been a growing literature, both theoretical and empirical, concerning parenting style in the context of child development. [Weinberg \(2001\)](#) and [Cosconati \(2009\)](#) present different frameworks to think about how parenting style decisions can augment children’s choices. [Weinberg \(2001\)](#) looks at two incentive schemes to motivate children, an allowance, or the threat of punishment, to show why authoritarian parenting practices may be more common among low-income parents. The model is static, and therefore cannot say anything about the non-cognitive evolution of children. [Cosconati \(2009\)](#) allows for parental limits to affect the marginal utility that children place on their leisure time, influencing their time allocation decisions, but again, their preferences and stock of non-cognitive skills are static as this work focuses on the parents learning their child’s type. [Akabayashi \(2006\)](#) looks at parental praise and punishment in a model of children’s human capital formation, and allows for complementarities in that a child’s patience is tied to their stock of cognitive skills. [Galiani et al. \(2017\)](#) presents a theoretical framework to explain the reduction in the use of punishment of children over the course of the 20th century. [Burton et al. \(2002\)](#) present a model of parental praise of a child’s good behavior, and show that children act out more often in poor households. The main takeaway is that child behavior is influenced by a range of socioeconomic factors, but, socioeconomic factors indirectly affect parenting practices through their response to child behavior. [Kim \(2019\)](#) analyzes the parent-child relationship using a signaling model, where the signaling method used by the parent is punishment. He shows that when the parent uses consistent punishment to signal poor behavior, the child will eventually be persuaded to behave optimally even when parents do not supervise their child.

The growing interest in the dynamics of the parent-child relationship is a natural next step in the child development literature. We have a sound understanding of the importance of parental time and monetary investment, and we know that the effectiveness of parental

time inputs decreases as children age². We also have work that shows as the effectiveness of parental time decreases, children’s own time investment becomes more productive (Del Boca et al. (2017)). This shift in the relative importance of parental and child time brings to light the importance of parenting style decisions that have dynamic implications. As parents who bring up children who place higher values on their cognitive development will not feel the need to exert effort to constrain their time allocation decisions.

There is also the opportunity to explore how we incorporate non-cognitive skills into models of the parent-child relationship. Thus far, those who have analyzed non-cognitive skills have tracked it as a stock, without allowing for a link to child behavior (Mullins (2022)). Seror (2022) allows for non-cognitive skills to influence children’s decisions. In his framework, children are trying to discern what the optimal action to take is, and higher non-cognitive skills lead to a higher value placed on minimizing the distance between their choice and the optimal action³. In this paper, children’s non-cognitive skills are mapped to their preferences, which influence their study time decisions, and children are forward looking, whereas they are myopic in Seror (2022). Del Boca et al. (2026) present a model where parenting style (a conditional cash transfer) is allowed to influence their child’s non-cognitive skill, which is summarized by their patience⁴. Their measure of patience from an external source, whereas this paper is perhaps the first to estimate a link between the Behavior Problems Index of the CDS and children’s preferences⁵.

While this paper presents a novel approach to think about parenting style and the development of children, like other works, elements of the decisions households make are not present in the framework. I do not consider the role of divorce present in Brown et al. (2025), nor do I allow for the parents to behave strategically with one another. The role of non-parental childcare that is considered by Bernal and Keane (2011), Griffen (2019), Brilli (2022), Verriest (2022), and Berlinski et al. (2024), is absent. The model presented below allows for parenting style decisions to interact with household time constraints, and therefore, it is likely that non-parental childcare choices, especially at early ages, could be linked to parenting style decisions. Further, while this paper considers an after school homework program as a counterfactual policy, parental work related policies such as those in Dahl and Lochner (2012), Griffen (2019), Mullins (2022), and Daruich (2024) are absent.

²See Del Boca et al. (2014) and Del Boca et al. (2026), Brilli (2022), and Verriest (2022) for this result in a framework similar to this paper

³See Lizzeri and Siniscalchi (2008) for an alternative framework on parental guidance in a model of parent-child interaction.

⁴See Kim (2026) for an alternative framework of a conditional cash transfer and child development.

⁵The effects of the different parenting styles and use corporal punishment on child development have been well documented in the sociology, psychology, and medical literatures, but have received limited attention in the field of economics, see Straus (1991), Baumrind (1991), Gershoff (2002), and Spera (2005)

Finally, the framework presented in this paper considers only household choices, and therefore, should be thought of as a partial equilibrium model. Here, parenting style decisions are driven by the child’s stock of non-cognitive skills, which are linked to their preferences and influence their study time choices, however, parenting style decisions are driven by more than just this one element of child behavior, and children’s cognitive and non-cognitive skills are a function of elements outside of the household. Parenting style decisions are also a function of the cultural norms in society as laid out by [Bisin and Verdier \(2001\)](#) and are also likely tied to network formation by children and their neighborhood and educational environment (see [Harris \(2024\)](#), [Agostinelli et al. \(2023\)](#), [Boucher et al. \(2022\)](#), and [Carvalho et al. \(2024\)](#)), a channel that is absent in this framework and thus, must be taken into account when interpreting the results concerning the effect of parenting style on the evolution of children’s non-cognitive skills.

The model in this paper is estimated using Simulated Method of Moments (SMM) with PSID-CDS data. The results show that the stock of children’s non-cognitive skills have a strong effect on their preference for leisure. Specifically, children who have low externalizing non-cognitive skills (that is, are more prone to behaviors such as acting out) place high (low) weight on their leisure time (cognitive skills). An interesting result is that children who have low internalizing non-cognitive skills (meaning they are more likely to display nervous/anxious behaviors) place a higher weight on their cognitive skills than children who have high internalizing skills. With respect to parenting style decisions, I find that, relative to a permissive parenting style, authoritarian and authoritative parenting have a negative effect on children’s externalizing non-cognitive skills, and a weakly positive effect on their internalizing non-cognitive skills. I show that parenting style decisions are impacted by a counterfactual policy that exogenously increases children’s study time, as more parents opt for a permissive parenting style. This policy raises the final stock of children’s cognitive skills, while their non-cognitive skills remain mostly unchanged.

The rest of the paper is organized as follows, Section [2](#) presents the theoretical model, Section [3](#) defines and summarizes the data used, Section [4](#) outlines the econometric assumptions made to estimate the model, Section [5](#) presents the arguments for identification of the structural parameters, Section [6](#) discusses the estimation of the structural model, Section [7](#) presents the model estimates, Section [8](#) discusses the model fit, Section [9](#) presents the results of the counterfactual policy, and Section [10](#) concludes.

2 Model

The model follows a set of households, each consisting of a mother, a father, and their child. In this setting, parents and the child make decisions that affect the cognitive and non-cognitive human capital development of the child within a Markov Perfect Equilibrium framework. Cognitive skills evolve in a manner similar to [Del Boca et al. \(2014\)](#) while the child's preferences are anchored to their non-cognitive skills, and are allowed to vary over time as a function of their parents' choice of a parenting style. Household choices are made in a leader-follower structure, with parents moving first, and their child second. Parents choose labor supply, how much time and money to invest in their child, and decide on a parenting style. Children observe the actions of their parents, and choose how much time to spend studying.

2.1 Environment

Each household consists of a mother (denoted by superscript m), a father (f), and a single child (c). Time is discrete and is indexed by the age of the child in years, denoted by t . Child skills are summarized by a three-dimensional vector

$$\mathcal{S}_t = \{k_t, B_t^e, B_t^n\}$$

where k_t represents their cognitive skills, B_t^e and B_t^n the externalizing and internalizing components of their non-cognitive skills.

Children's have preferences over their leisure time, l_t^c , and the stock of their cognitive skills. A novel feature of this paper is that children's preferences are allowed to be a function of their stock of non-cognitive skills, and thus, are time-varying and endogenous. Their preferences are represented by:

$$u_t^c(l_t^c, k_t, B_t^e, B_t^n) = \omega(B_t^e, B_t^n) \ln(l_t^c) + (1 - \omega(B_t^e, B_t^n)) \ln(k_t) \quad (1)$$

Children's preferences are Cobb-Douglas, and ω maps the stock of non-cognitive skills to the child's preferences.

Parents operate under a unitary model and are altruistic toward their child, their utility function is:

$$\tilde{u}_t^p = (1 - \varphi) u_t^p(l_t^m, l_t^f, c_t, k_t, ps_t) + \varphi u_t^c(l_t^c, k_t, B_t^n, B_t^e) \quad (2)$$

where φ governs parental altruism, l_t^m is the mother's leisure time, l_t^f the father's, c_t is parental consumption, and ps_t is the parents' choice of a parenting style. The private com-

ponent of the parents' utility function is:

$$u_t^p(l_t^m, l_t^f, c_t, k_t, ps_t) = \alpha_l^m \ln(l_t^m) + \alpha_l^f \ln(l_t^f) + \alpha_c \ln(c_t) + \alpha_k \ln(k_t) - \xi(ps_t, \tau_t^c) \quad (3)$$

where $\xi(ps_t, \tau_t^c)$ is the parents' dis-utility from their choice of parenting style, which is allowed to be a function of child study time. Equation (2) allows parent utility to directly depend on their child's stock of cognitive skills, and indirectly on the stock of non-cognitive skills due to their influence on their child's preference for leisure, which will influence their time allocation decision.

The assumption of a unitary model rules out strategic interaction between parents when making labor supply and child investment decisions. It is also implied that the choice of parenting style is at the household level as assumed by [Cosconati \(2009\)](#) and [Agostinelli et al. \(2023\)](#) (who both consider a child and a representative parent, therefore parenting style is automatically at the household level), ruling out different parenting style decisions by each parent. This decision is driven by data limitations, as I am typically missing information about the father's parenting style, however, when data is available for both parents, the majority of the time, the parents agree on a style. While the unitary model has its limitations, it is a common assumption in the literature (see [Weinberg \(2001\)](#), [Akabayashi \(2006\)](#), [Bernal \(2008\)](#), [Hao et al. \(2008\)](#), [Liu et al. \(2010\)](#), [Del Boca et al. \(2014\)](#), and [Berlinski et al. \(2024\)](#), etc.)

Each period, parents jointly make a parenting style decision, $ps_t \in \{AR, AT, P\}$, where AR denotes an authoritarian parenting style, AT an authoritative parenting style, and P a permissive parenting style⁶. They also make continuous choices regarding labor supply, monetary investment in their child, and time spent with their child subject to a household budget constraint and individual time constraints. The household budget constraint is given by:

$$w_t^f h_t^f + w_t^m h_t^m + y_t = c_t + g_t \quad (4)$$

where h_t^j represents parent j's hours worked in the labor market at hourly wage w_t^j , y_t is non-labor income and g_t are monetary investments in their child, or child goods⁷. Each parent faces a time constraint

$$112 = \tau_t^j + h_t^j + l_t^j \quad (5)$$

where τ_t^j is the time that parent j spends with their child. It is assumed that each parent has

⁶For a detailed explanation of the three major parenting styles, see [Baumrind \(1966\)](#). For an introduction to the economics of parenting style decisions, see [Doecke and Zilibotti \(2017\)](#).

⁷Unlike [Caucutt et al. \(2026\)](#), there are no capital markets, everything that is earned is either consumed by the parents or invested in the form of child goods. This is a common assumption for those who have followed the [Del Boca et al. \(2014\)](#) framework

112 hours each week to allocate toward work, leisure, and time with their child, therefore, all time investment decisions are in hours per week. This implies that total household income, consumption and expenditures on children are also at the weekly level.

The child only faces a time constraint, which is given by

$$112 = \tau_t^c + l_t^c + s_t + \tau_t^m + \tau_t^f \quad (6)$$

where τ_t^c are the number of hours per week the child spends studying and s_t is exogenous time spent in school. Note that the time parents spend with their child appears in the child's time constraint, effectively setting an upper bound for the maximum amount of time a child can spend studying, and therefore, can crowd out child study time.

Children's cognitive and non-cognitive skills are produced as a function of parental and child decisions. The production function for children's cognitive skills is:

$$\ln(k_{t+1}) = \ln(R_t) + \delta_t^m \ln(\tau_t^m) + \delta_t^f \ln(\tau_t^f) + \delta_t^c \ln(\tau_t^c) + \delta_t^g \ln(g_t) + \delta_t^k \ln(k_t) \quad (7)$$

where R_t is total factor productivity.

Non-cognitive skills, B_t^n and B_t^e are each binary random variables that are a function of the child's age, and the parents' choice of a parenting style. Their laws of motion are:

$$B_{t+1}^n = f_t^n(t, ps_t, B_t^e, B_t^n) \quad (8)$$

and

$$B_{t+1}^e = f_t^e(t, ps_t, B_t^e, B_t^n) \quad (9)$$

The reader may note that while non-cognitive skills do not depend on cognitive skills (consistent with recent work by [Del Boca et al. \(2026\)](#)), cognitive skills are indirectly influenced by non-cognitive skills through children's preferences, which affect the child's time allocation decision, and therefore, the parents' time investment and parenting style choices.

The parents and the child are both forward looking agents who differ with respect to about how they think about the future. The state vector in any period is:

$$\Gamma_t = (w_t^f, w_t^m, y_t, k_t, B_t^e, B_t^n)$$

The value function for the parents, who anticipate the actions of their child is

$$V_t^p(\Gamma_t) = \max_{a_t^p | \tau_t^c(a_t^p)} \tilde{u}_t^p + \beta^p E_t V_{t+1}^p(\Gamma_{t+1} | a_t^p, \Gamma_t) \quad (10)$$

where β^p is the parents' discount factor and $a_t^p = (\tau_t^m, \tau_t^f, h_t^m, h_t^f, ps_t, g_t)$ is their choice vector.

Children know the law of motion for their cognitive skills, but are slightly myopic because they assume their preference for leisure (or more specifically their stocks of non-cognitive skills) will remain unchanged in the future, an assumption the parents are aware of.⁸ Their value function is given by:

$$V_t^c(\Gamma_t) = \max_{\tau_t^c | a_t^p} u_t^c(l_t^c, k_t, B_t^n, B_t^e) + \beta^c E_t V_{t+1}^c(\tilde{\Gamma}_{t+1} | \tau_t^c, a_t^p, \Gamma_t) \quad (11)$$

where $\tilde{\Gamma}_{t+1}$ accounts for the assumption that children do not believe their non-cognitive skills will evolve.

The difference in preferences concerning children's skills is a factor that drives parenting style decisions. Typically, children place a smaller weight on their cognitive skills than their parents would desire. This source of conflict could cause parents to be more likely to adopt a parenting style that encourages effort that their child would not put forth on their own accord. Additionally, parenting style decisions today influence a child's non-cognitive skills, and therefore, their preferences, tomorrow, so parents may also choose a parenting style to mold their child's preferences to be closer to those desired by the parents.

Children, unaware that their preferences are malleable, only place a terminal value on the final stock of their cognitive skills:

$$V_{T+1}^c(k_{T+1}) = \psi^c(1 - \omega(B_T^e, B_T^n)) \ln(k_{T+1}) \quad (12)$$

Parents care not only about the final stock of their child's cognitive skills, but their non-cognitive skills as well, as they would like to develop a child who places a higher weight on their own cognitive skills. Their terminal value is:

$$V_{T+1}^p(k_{T+1}) = \psi^p((1 - E_T \omega(B_{T+1}^e, B_{T+1}^n)) + \alpha_k) \ln(k_{T+1}) \quad (13)$$

where the expectation operator is taken over the child's future stock of non-cognitive skills and ψ^c and ψ^p are positive parameters to be estimated that are common among all parents and children.

After the child is T years old, the child development process outlined in this model is no longer relevant, and the child enters a different stage of their lifecycle. It can be assumed that both the parents' and the child solve different optimization problems than the one described above, and the child's skill production functions changes as well. Therefore, these values can be thought of as initial conditions in the next phase of life for both the parents and the

⁸This follows from Akabayashi (2006).

child.

2.2 Parenting Style

Prior to describing the model solution, I must distinguish how the different parenting styles, authoritarian, authoritative, and permissive, interact in the model. As seen in (8) and (9) the choice of parenting style affects the development of children's future preferences. In this framework, a parenting style can be classified as intensive or non-intensive (Doepke and Zilibotti (2017)). Intensive parenting in this setting implies that parents set rules or limits concerning how their child spends their time. A non-intensive parenting style in this framework is a permissive parenting style. If parents choose to be permissive, then children are free to make their study time decision on their own accord. If the parents choose an intensive parenting style, they restrict the choice of their child. Intensive parenting can take one of two forms, parents can either be warm or cold, distinguishing authoritarian (cold), and authoritative (warm) parenting (Baumrind (1991)).

2.3 Household Equilibrium

Each period, the parents and their child interact in a leader-follower structure where the parents move first and choose $(\tau_t^m, \tau_t^f, h_t^m, h_t^f, ps_t, g_t)$, and, conditional on the parents' actions, the child chooses their study time τ_t^c . An added wrinkle is that children only really choose their study time if their parents choose a permissive parenting style. If the parents choose one of the intensive parenting styles (authoritarian or authoritative), the parents constrain the choice of their child.

2.3.1 Case 1: Parents Are Permissive

In this case, children follow the action of their parents, and make an unconstrained choice regarding their study time. In any period t , the child solves

$$V_t^c(\Gamma_t) = \max_{\tau_t^c | a_t^p} u_t^c(l_t^c, k_t, B_t^e, B_t^n) + \beta^c E_t V_{t+1}^c(\tilde{\Gamma}_{t+1} | \tau_t^c, a_t^p, \Gamma_t)$$

subject to their time constraint

$$112 = \tau_t^m + \tau_t^f + l_t^c + \tau_t^c + s_t$$

The solution to the child's period t problem is given by⁹

$$\tau_t^c(ps_t = P) = \frac{\beta^c D_{t+1}^c \delta_t^c}{\omega(B_t^e, B_t^n) + \beta^c D_{t+1}^c \delta_t^c} (112 - \tau_t^f - \tau_t^m - s_t) \quad (14)$$

where

$$D_t^c = \frac{\partial E_t V_t^c}{\partial \ln(k_t)}$$

represents the marginal utility the child places on their (log) cognitive skills next period. The child's reaction function has the natural properties in that it is increasing in the marginal productivity of their study time and decreasing in parental time investment. The more interesting aspect of the child's reaction function is that their study time is a function of their stock of non-cognitive skills, B_t^n and B_t^e , which can raise or reduce $\omega(B_t^e, B_t^n)$, influencing study time.

The parents' choose $a_t^p = (\tau_t^m, \tau_t^f, h_t^m, h_t^f, g_t, ps_t)$ to maximize

$$V_t^p(\Gamma_t) = \max_{a_t^p | \tau_t^c(a_t^p)} \tilde{u}_t^p + \beta^p E_t V_{t+1}^p(\Gamma_{t+1} | a_t^p, \Gamma_t)$$

subject to

$$w_t^f h_t^f + w_t^m h_t^m + y_t = c_t + g_t$$

their individual time constraints

$$112 = \tau_t^j + h_t^j + l_t^j$$

for $j \in \{m, f\}$, and must also take into account the child's time constraint, the law of motion for cognitive skills, and the effect of their choice of parenting style on their child's future non-cognitive skills.

Let,

$$D_{t+1}^p = \frac{\partial E_t [V_{t+1}^p | \Gamma_t, ps_t]}{\partial k_{t+1}}$$

represent the marginal continuation value parents place on their child's cognitive skills in period $t+1$, which accounts for the fact that their choice of a parenting style will influence next period's non-cognitive skills, and therefore, decisions made next period.

The solution to the parents' problem requires a numerical solution for τ_t^m and τ_t^f , however, conditional on these solutions, I am able to derive closed for solutions for labor supply and

⁹A full derivation of household equilibrium can be found in Appendix A

goods investment. Conditional on ps_t , τ_t^m , and τ_t^f , parent labor supply is

$$h_t^f(\tau_t^m, \tau_t^f) = \frac{w_t^f(112 - \tau_t^f)((1 - \varphi)(\alpha_l^m + \alpha_c) + \beta^p D_{t+1}^p \delta_t^g) - (1 - \varphi)\alpha_l^f(w_t^m(112 - \tau_t^m) + y_t)}{w_t^f((1 - \varphi)(\alpha_l^f + \alpha_l^m + \alpha_c) + \beta^p D_{t+1}^p \delta_t^g)} \quad (15)$$

and

$$h_t^m(\tau_t^m, \tau_t^f) = \frac{w_t^m(112 - \tau_t^m)((1 - \varphi)(\alpha_l^f + \alpha_c) + \beta^p D_{t+1}^p \delta_t^g) - (1 - \varphi)\alpha_l^m(w_t^f(112 - \tau_t^f) + y_t)}{w_t^m((1 - \varphi)(\alpha_l^f + \alpha_l^m + \alpha_c) + \beta^p D_{t+1}^p \delta_t^g)} \quad (16)$$

where the observed choice is

$$h_t^{j,*}(\tau_t^m, \tau_t^f) = \max\{h_t^j(\tau_t^m, \tau_t^f), 0\}$$

With these choices in hand, the solution for child goods, g_t , is a fraction of total household income

$$g_t(\tau_t^m, \tau_t^f) = \frac{\beta^p D_{t+1}^p \delta_t^g}{(1 - \varphi)\alpha_c + \beta^p D_{t+1}^p \delta_t^g} (w_t^f h_t^{f,*}(\tau_t^m, \tau_t^f) + w_t^m h_t^{m,*}(\tau_t^m, \tau_t^f) + y_t) \quad (17)$$

2.3.2 Case 2: Authoritarian & Authoritative Parenting

In this case, parents use an intensive parenting style to constrain their child's choice of study time. Prior to laying out the solution, a discussion on the parenting style cost function $\xi(ps_t, \tau_t^c)$ is necessary. Permissive parents do not incur a monitoring cost because they do not set limits for their child and therefore have no monitoring responsibilities. In the permissive case $\xi(P) = 0$. Intensive parenting requires setting and monitoring limits, which comes with a monitoring cost. This cost is defined as

$$\xi(ps_t, \tau_t^c) = \xi^{ps} \ln(\tau_t^c)$$

for $ps \in \{AR, AT\}$, parents incur a marginal cost to set a higher study time limit. In this case, children do nothing and parents make all of the choices in the household. Conditional on a pair (τ_t^m, τ_t^f) , the first order condition for child study time chosen by the parents is:

$$-\frac{(1 - \varphi)\xi^{ps}}{\tau_t^c} - \frac{\varphi\omega(B_t^c, B_t^n)}{\tilde{T} - \tau_t^c} + \frac{\beta^p D_{t+1}^p \delta_t^c}{\tau_t^c} = 0 \quad (18)$$

The first term of (18) represents the marginal cost of monitoring, the second term reflects that by setting a higher study time limit, the parents reduce their child's utility, because

they are altruistic, this lowers parent utility.¹⁰ The final term is the marginal benefit of increased study time, which increases children’s cognitive skills (and also takes into account the expected impact of the parenting style on next period non-cognitive skills, which is captured in D_{t+1}^p).

The solution for study time set by the parents is

$$\tau_t^c = \frac{\beta^p D_{t+1}^p \delta_t^c - (1 - \varphi) \xi^{ps}}{\varphi \omega(B_t^n, B_t^e) + \beta^p D_{t+1}^p \delta_t^c - (1 - \varphi) \xi^p} \tilde{T} \quad (19)$$

which is of course constrained to be non-negative. With $(\tau_t^m, \tau_t^f, \tau_t^c)$ all in hand, the solutions for h_t^m , h_t^f , and g_t are computed as seen in the previous section.

2.3.3 Optimal Parenting Style

Each parenting style $ps_t \in \{AR, AT, P\}$ generates a unique choice of $\{\tau_t^m, \tau_t^f, h_t^m, h_t^f, \tau_t^c, g_t\}$ that is associated with a unique value, $V_t^p(ps_t)$ for the parents. The optimal parenting style choice is then the parenting style that generates the largest value for the parents. When making their parenting style decision, parents consider their own private dis-utility, the utility of their child, future cognitive skills, and how their choice of a parenting style will affect their child’s future non-cognitive skills.

3 Data and Descriptive Statistics

The data used in this paper are taken from the Panel Study of Income Dynamics (PSID) and its Child Development Supplement (CDS). The PSID is a dynastic survey of households in the United States that began in 1968 with a set of approximately 18000 individual from about 5000 families. Between 1968 and 1997, the PSID surveyed households annually, incorporating new members as the original families expanded, since 1997, the PSID has been collected bi-annually. Starting the 1997, the CDS collected information on 3563 children living in 2394 PSID families with children between the age of 3 and 13 years old. The CDS was collected in three waves, the first, CDS-I, was collected in 1997, CDS-II was collected in 2002, and CDS-III in 2007, no new families were added in the second and third waves. The PSID-CDS is well-suited for this study because it contains information about parental labor supply and income, has data on parent and child time use, as well as information about children’s cognitive and non-cognitive skills.

¹⁰This is a key difference between this paper and the work in [Del Boca et al. \(2026\)](#). In their case, parents could offset the reduction in child leisure time by paying them an allowance, this channel is absent in this framework, an increase in study time simply reduces child utility.

The main PSID survey provides information on parent demographic data such as age and education, as well as labor supply, hourly wages, and data on non-labor income. I collect this information for parents from the year their child was 4 years old through the year that their child turns 18, providing a panel from 1986 to 2009. CDS-I, CDS-II, and CDS-III provides data on children’s cognitive and non-cognitive skills, as well as parent and child time allocation, and information about parenting style decisions. To construct my sample, I follow [Del Boca et al. \(2026\)](#) and require that children have a cognitive and non-cognitive skill measure in 1997 and 2002, their biological parents are married when their child is between 4 and 18 years old, and require that the child either have no siblings, or, if there is a household with two children, the younger child is selected. This resulting sample has information on 270 children and their parents¹¹.

3.1 Systemically Missing Data

The main PSID has been conducted bi-annually with data corresponding to the previous year starting in 1997, prior to this, it was conducted annually. The CDS was conducted in three waves, 1997, 2002, and 2007, therefore, a motivation for using simulation methods (described below) is due to systemically missing data in the PSID-CDS. Table 1 shows data availability for a PSID-CDS household with a child that was born in 1993. The only year where the PSID and CDS surveys overlap is 2002, the typical case is that parental wage and labor supply data is available or, data from the CDS is available.

Table 1. **Structure of Data**

Data	Years Available	Source
Skill Measures	1997, 2002, 2007	CDS
Parent Time	1997, 2002, 2007	CDS
Parenting Style	1997, 2002, 2007	CDS
Child Study Time	1997, 2002, 2007	CDS
Parent Labor Supply	1993-1996, 1998-2009 (bi-annually)	PSID
Parent Wages	1993-1996, 1998-2009 (bi-annually)	PSID
Household Non-Labor Income	1993-1996, 1998-2009 (bi-annually)	PSID

Notes: The PSID-CDS data exhibits non-random missing data due to the survey structure. This table presents a sample of data availability for a PSID-CDS household with a child born in 1993.

¹¹For comparison, [Del Boca et al. \(2014\)](#) have 105 children in their sample, [Del Boca et al. \(2026\)](#) have 248, and [Verriest \(2022\)](#) has 247. All use the PSID-CDS and make similar sample selection decisions over different age ranges.

Summary Statistics

Table 2. **Summary Statistics**

	Mean	St Dev	Min	Max
LW Score	41.39	13.87	4.00	57.00
Externalizing Behavior (BPIE)	0.66	0.47	0.00	1.00
Internalizing Behavior (BPIN)	0.62	0.49	0.00	1.00
Mother Time with Child	16.51	12.74	0.00	70.25
Father Time with Child	7.64	10.60	0.00	101.25
Mother Time if Authoritarian	14.52	11.56	0.00	51.00
Mother if Authoritative	16.71	13.02	0.00	65.00
Mother Time if Permissive	17.51	13.18	0.00	70.25
Father Time if Authoritarian	7.25	9.26	0.00	38.50
Father if Authoritative	7.53	11.73	0.00	101.25
Father Time if Permissive	7.11	8.93	0.00	40.00
Child Study Time	4.17	6.29	0.00	37.42
Child Study Time BPIE = 1	4.28	6.33	0.00	37.42
Child Study Time BPIE = 0	3.94	6.19	0.00	33.00
Child Study Time BPIN = 1	3.86	5.72	0.00	32.33
Child Study Time BPIN = 0	4.67	7.09	0.00	37.42
Child Study Time Authoritarian	4.37	6.55	0.00	30.23
Child Study Time Authoritative	4.37	6.37	0.00	37.42
Child Study Time Permissive	4.48	6.61	0.00	32.33
Frac Authoritarian	0.27	0.45	0.00	1.00
Frac Authoritative	0.46	0.50	0.00	1.00
Frac Permissive	0.27	0.44	0.00	1.00
Frac Mothers Work	0.84	0.37	0.00	1.00
Mother's Work Hours	30.40	13.04	0.08	80.00
Frac Fathers Work	0.98	0.14	0.00	1.00
Father's Work Hours	43.64	9.75	0.81	80.00
Mother's Wage	18.17	12.87	0.00	134.94
Father's Wage	28.13	21.46	0.00	149.61
Non-Labor Income	150.05	205.42	0.00	980.38

Notes: Summary statistics are based on the author's sample of 270 households from the PSID-CDS.

Table (2) above presents descriptive statistics for the variables used in this paper. Consistent with others who have used the PSID-CDS data, on average mother's spend more active time with their child than fathers, spending on average 16.51 hours per week with their child compared to fathers who spend on average 7.64 hours per week. Children spend about 4.17 hours per week studying, and this is not conditional on the parenting style that they face. Average child study time is about the same when conditioning on the parenting style

parents adopt, with average study time being about 4.37 hours per week when children face an authoritarian or authoritative parenting style, and children studying slightly more when facing a permissive parenting style (4.48 hours per week)¹². Permissive parenting is characterized by allowing children to make decisions with minimal interference, therefore, it is understandable that children who face a permissive parenting style likely will study enough on their own without any parental involvement, which might be why study time conditional on permissive parenting is slightly higher than authoritarian or authoritative parenting.

There are also differences in parental time investment choices conditional on the household's choice of a parenting style. Mothers on average spend the most time with their children conditional on a permissive parenting style, and the least amount of time with their children conditional on an authoritarian parenting style. The spread for fathers is tighter, however, fathers spend the most time on average with their child when the household chooses an authoritative parenting style, and the least when the household chooses a permissive parenting style¹³.

There is also variation in child study time conditional on the child's stocks of non-cognitive skills. On average, children who have high externalizing non-cognitive skills (BPIE equal to 1) study 4.28 hours per week, where this amount is 3.94 hours per week for those with low externalizing skills. Children with high internalizing non-cognitive skills study less than those with low internalizing non-cognitive skills on average, which might be because children who have low internalizing skills are more anxious, and these anxieties may be tied to academic performance, which could increase study time.

The pattern for parental labor supply follows the trends observed in the literature. Fathers on average supply labor more often than mothers (0.98 compared to 0.84), and, conditional on working, work longer hours (43.64 hours per week versus 30.40 hours per week) at a higher wage (\$28.13 per hour versus \$18.17 an hour).

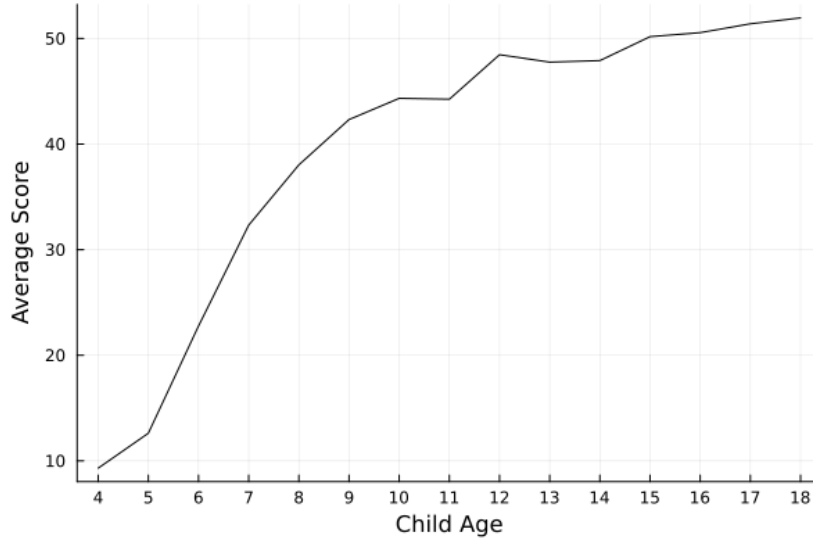
3.2 Cognitive and Non-Cognitive Skills

I measure children's cognitive skills (k_t) with the Letter-Word (LW) module of the Woodcock-Johnson aptitude test. This assesses children's reading and symbolic identification skills, and has the advantageous property in that it is administered to children in the CDS between the ages of 3 and 18, allowing me to track the development of skills over time. Figure 1 plots average child performance for my sample.

¹²Unconditional average study time will differ from study time conditional on a parenting style because of the inclusion of study time when no parenting style data is available.

¹³In this paper, parenting style is at the household level and reflects the Primary Caregiver survey in the CDS, which is almost always answered by the mother. Therefore, when thinking about father's time investment, it is conditional on the mother's choice of a parenting style.

Figure 1. **Average Child Skills By Age**



Notes: Average child performance on the LW Exam in the author’s sub-sample of the CDS. The exam is out of 57 questions.

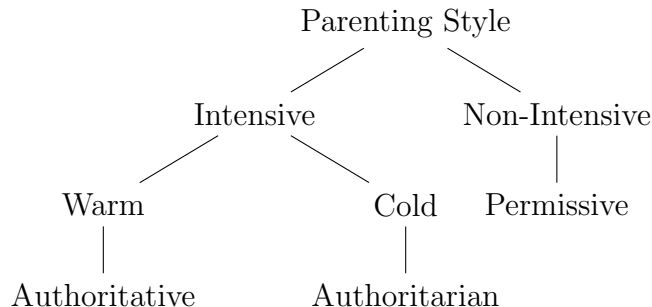
The measure I use for children’s non-cognitive skills are the externalizing and internalizing components of the Behavior Problem Index (BPI). Each wave, the CDS interviews the child’s primary caregiver (PCG), and asks them if their child exhibits certain behavior issues. The PCG can respond with “Never”, “Sometimes”, and “Often”, the responses are then summed up to yield a total score. In my application, I reverse the scoring such that a higher score corresponds to a child with fewer behavior issues and therefore higher non-cognitive skills. I then divide the BPI scores into two bins, those above and below the median scores. Non-cognitive skills are binary random variables where B_t^n and B_t^e can take on values $\{0, 1\}$, where 1 indicates a “high” level of the non-cognitive skill and 0 indicates “low” non-cognitive skill.

3.3 Parenting Style Classification

A question asking parents if they implement an authoritarian, authoritative, or permissive parenting style is not available in the CDS data. Instead, I must use other questions available to construct a measure of parenting style. As described in Section 2.2, we can think about parenting styles along two dimensions, obedience/control, and warmth (Maccoby and Martin (1983) and Baumrind (1991)). Permissive parenting is distinct from authoritarian and authoritative parenting in that it is non-intensive (Doepke and Zilibotti (2017) and Doepke et al. (2019)). The intensive parenting styles are distinct from one another depending on the level of warmth parents provide their child, intensive and warm parenting is authoritative, whereas intensive and cold is authoritarian. An illustration of the parenting style

classification is shown in Figure 2.

Figure 2. Depiction of Parenting Styles



To determine whether parents are intensive or not, I follow [Doepke and Zilibotti \(2017\)](#) and [Doepke and Zilibotti \(2019\)](#). The CDS asks the primary caregiver (PCG) “If you had to choose, which thing on this list would you pick as most important for a child to learn, to prepare him or her for life?” Potential responses are: “To obey”, “To be well-liked or popular”, “To think for himself/herself”, “To work hard”, “To help others when they need help”. The primary care-giver is then asked this same question, but for their second, third, and fourth most important quality, the potential answers are the same. Parents are intensive if their first or second values are “To obey” or “To work hard”, if this is not the case, parents are classified as having chosen a permissive parenting style.

Distinguishing between authoritarian and authoritative parenting in data is more difficult. To do so, I use the measure of parental warmth available in the CDS. Each wave, the PCG is asked a series of questions regarding physical affection, saying “I love you”, participation in activities, playing or joking, talking about interests, and speaking appreciatively with their child. These scores are then summarized into a parental warmth scale, which I then standardize. Of the parents who are classified as having chosen an intensive parenting style, if the standardized score is above 0, they have chosen an authoritative style, if it is below 0, they have chosen an authoritarian parenting style.

Table 2 shows that in the sample, authoritative parenting is chosen most often, and permissive and authoritarian parenting are chosen with about equal frequency. This is consistent with work by [Majumder \(2016\)](#) also report that authoritative parenting is chosen most often in U.S households, albeit they use different data and a different methodology to classify parenting styles. One may have concerns that I allow for parenting style to be a choice in my framework and not a type. Table 3 shows that parenting style choices change in the data, providing evidence that parenting styles should be considered a choice, and not

a type.

Table 3. **Parenting Style Transitions**

	Authoritarian	Authoritative	Permissive
Authoritarian	0.53	0.25	0.21
Authoritative	0.19	0.57	0.23
Permissive	0.21	0.35	0.44

Notes: Data is taken from the author’s subsample of the CDS. Each unit of observation is a household when a child is t years old, and a household when the child is $t+5$ years old.

Parenting style is also correlated with demographic characteristics, including education. Table 4 shows that an authoritarian parenting style is adopted more often in households where the mother has at most a high school degree. This is an empirical fact consistent with the literature that parenting style choices are correlated with socioeconomic status (see Doepke and Zilibotti (2017), Cobb-Clark et al. (2019), and Harris (2024) for example).

Table 4. **Parenting Style By Mother’s Education**

Parenting Style	Some College	High School
Authoritarian	0.26	0.31
Authoritative	0.46	0.44
Permissive	0.28	0.25

Notes: Data is taken from the author’s subsample of the CDS. College represents households where the mother has a college degree (or more), No College represents households where the mother does not have a college degree.

4 Econometric Specifications

To map the model to the data, I make the following econometric specifications with respect to household preferences, the child skill development production functions, the wage offer distribution, and the non-labor income process.

4.1 Preferences

The parameters governing parental preferences over their leisure time (α_l^m and α_l^f), consumption (α_c), and child cognitive skills (α_k) are assumed to be constant across time, and common to all households. These parameters are constrained to be positive and sum to one.

Parenting style costs, $\xi(ps_t, \tau_t^c)$, are allowed to vary across households, but are constant over time. The cost of permissive parenting, $\xi(P)$ is normalized to 0. Authoritarian and

authoritative parenting are both intensive parenting styles that are characterized by enforcing rules and limits on their child¹⁴. Intensive parenting is then associated with a monitoring cost that is allowed to vary across parents. Authoritative parenting is distinct from authoritarian because it is associated with parental warmth, therefore, parents choose to exert extra effort to be warm toward their children while monitoring their study time. The marginal costs of authoritarian and authoritative parenting are then:

$$\xi(AR) = \exp(\kappa^i) \quad (20)$$

$$\xi(AT) = \exp(\kappa^i + \kappa^w) \quad (21)$$

where

$$\kappa^i = \mu^i + \epsilon^i$$

and

$$\epsilon^i \sim N(0, \sigma^i)$$

Finally, $\kappa^w > 0$, the cost of warm monitoring, is common to all parents¹⁵.

A unique feature of this paper is that children's preferences are allowed to vary over time as a function of their parents' choice of parenting style, and depend on their internalizing (B_t^n) and externalizing (B_t^e) non-cognitive skills. Children's preferences over leisure and their cognitive skills are constrained to be positive and sum to one. Their preference for leisure in any period t is

$$\omega(B_t^n, B_t^e) = \frac{\exp(\omega_0 + \omega_e B_t^e + \omega_n B_t^n + \omega_{en} B_t^e B_t^n)}{1 + \exp(\omega_0 + \omega_e B_t^e + \omega_n B_t^n + \omega_{en} B_t^e B_t^n)} \quad (22)$$

where $(\omega_0, \omega_e, \omega_n, \omega_{en})$ are parameters to be estimated. This implies that children's preference for leisure can take on one of four values, depending on whether or not they have high or low externalizing and internalizing non-cognitive skills. I do not constrain the parameters such that higher levels of non-cognitive skills produce lower preferences for leisure (and therefore place higher weight on their cognitive skills), and let the data determine the sign of these parameters. The child's discount factor is also exogenously set to for all children.

¹⁴While these are traits of authoritarian and authoritative parenting, I do not model individual rule choices parents make, such as monitoring study time or screen time. See [Tracey et al. \(2024\)](#) and [Hotz and Pantano \(2015\)](#) for a discussions concerning children's human capital and individual parental rules.

¹⁵This rules out that it can become relatively more difficult, or easier, to implement a certain parenting style as children age. Authoritarian parenting is a demanding and controlling parenting style, as children age, they may rebel more against this style, causing friction and strain within the household ([Sarwar \(2016\)](#)).

4.2 Skill Production Function Parameters

I allow the parameters governing the production of child skills to vary monotonically with the age of the child¹⁶. The marginal product of parental and child time investment is allowed to be a function of parental education. The functional forms for parental time investment are:

$$\delta_t^j = \exp(\gamma_0^j + \gamma_1^j t + \gamma_2^j E^j) \text{ for } j \in \{m, f\}, \text{ for } t \in \{1, \dots, T\} \quad (23)$$

where E^j is parent j 's level of education in years. The marginal product of child study time is

$$\delta_t^c = \exp(\gamma_0^c + \gamma_1^c t + \gamma_2^c E^m) \text{ for } t \in \{1, \dots, T\} \quad (24)$$

where the mother's level of education is allowed to influence the marginal product of child study time. The marginal product for lagged cognitive skills and monetary investments take the form:

$$\delta_t^q = \exp(\gamma_0^q + \gamma_1^q t) \text{ for } q \in \{k, g\}, \text{ for } t \in \{1, \dots, T\} \quad (25)$$

Finally, TFP takes a logistic form that follows [Del Boca et al. \(2026\)](#)

$$\delta_t^R = \delta_0^R + \frac{\delta_1^R - \delta_0^R}{1 + \exp(-\delta_3^R(t - \delta_4^R))} \text{ for } t \in \{1, \dots, T\} \quad (26)$$

4.3 Non-Cognitive Skill Production

A child's externalizing (B_t^e) and internalizing (B_t^n) components of their non-cognitive skills are discrete random variables that are allowed to be influenced by their age, the current stock of their non-cognitive skills and their parents' choice of parenting style. The functional form for the probability a child has high non-cognitive skill N in period $t+1$ is

$$p_{t+1}^N = \frac{\exp(\beta_0^N + \beta_1^N t + \beta_2^N (ps_t = AR) + \beta_3^N (ps_t = AT) + \beta_4^N B_t^e + \beta_5^N B_t^n)}{1 + \exp(\beta_0^N + \beta_1^N t + \beta_2^N (ps_t = AR) + \beta_3^N (ps_t = AT) + \beta_4^N B_t^e + \beta_5^N B_t^n)} \quad (27)$$

for $N \in \{e, n\}$. This implies each skill B_t^N is a Bernoulli random variable with probability p_t^N .

4.4 Wages and Non-Labor Income

The log-wage offer distribution for parents is assumed to be only a function of parent characteristics and is not history dependent. This is a common assumption of those who use

¹⁶This is a common assumption in the literature, see [Del Boca et al. \(2014\)](#), [Brilli \(2022\)](#), and [Del Boca et al. \(2026\)](#).

the PSID-CDS data in the context of child development, and it reduces the computational burden of the solution to the household's problem¹⁷. Parent wages are assumed to be a function of their age and education, and log-wage offers are drawn each period from the following distribution:

$$\begin{pmatrix} \ln(w_t^f) \\ \ln(w_t^m) \end{pmatrix} \sim Normal \left(\begin{pmatrix} \eta_t^f \\ \eta_t^m \end{pmatrix}, \begin{pmatrix} \sigma_f^2 & \rho\sigma^w \\ \rho\sigma^w & \sigma_m^2 \end{pmatrix} \right) \quad (28)$$

where

$$\eta_t^j = \eta_0^j + \eta_1^j A_t^j + \eta_3^j E^j \quad (29)$$

for $j \in \{m, f\}$, A_t^j represents parent j 's age when the child is t years old, and E^j is parent j 's years of schooling.

I follow Verriest (2022) for the non-labor income process, which is assumed to only be a function of the father's demographic characteristics. Non-labor income is drawn using a two-step procedure. First, I draw a Bernoulli random variable to determine whether or not a household has a positive amount of non-labor income in a given period. The probability that a household has any positive non-labor income when their child is t years old is given by

$$Pr(y_t > 0) = \frac{\exp(\xi_{y_t})}{1 + \exp(\xi_{y_t})} \quad (30)$$

where

$$\xi_{y_t} = \mu_0^y + \mu_1^y A_t^f + \mu_2^y (A_t^f)^2 + \mu_3^y E^f \quad (31)$$

Conditional on a household having positive non-labor income in a given period, the level of non-labor income is drawn according to

$$y_t = \exp(\mu_4^y + \mu_5^y A_t^f + \mu_6^y (A_t^f)^2 + \mu_7^y E^f + \epsilon_t^y) \quad (32)$$

where $\epsilon_t^y \sim N(0, \sigma_y^2)$.

4.5 Measurement of Child Cognitive Skills

I assume that child skills, k_t , are observed by all members of the household, but the econometrician only has access to a noisy measure of each skill, the LW exam. In the model, child skills are a continuous variable, however, in the data, the noisy measures take on discrete values. To derive a mapping from latent child skills to the measures I observe in the data, I

¹⁷This same assumption is made by Del Boca et al. (2014), Brilli (2022), and Del Boca et al. (2026). For an example with a relaxation of this assumption, check out Verriest (2022).

follow [Lord and Novick \(1968\)](#). As a matter of notation, I refer to the latent values of child skills as k_t and k_t^* represent the measures of these skills observed in the CDS.

In the data, cognitive skills are measured by the total number of questions a child answers on the LW exam. There are 57 questions on the LW exam and, as in [Del Boca et al. \(2014\)](#) I make the assumption that each question has an equal level of difficulty and the probability a child answers a question is a function of their cognitive skills k_t . Let the probability a child with cognitive ability k_t answers a question on the LW exam correctly be given by

$$p(k_t) = \frac{k_t}{1 + k_t} \quad (33)$$

This assumption has attractive properties such that the probability a question is answered correctly is increasing in child skills, and is bounded between 0 and 1. The general form for the probability is given by

$$p(k_t, \lambda) = \frac{\exp(\lambda_0 + \lambda_1 \ln(k_t))}{1 + \exp(\lambda_0 + \lambda_1 \ln(k_t))} \quad (34)$$

however, as with all factor models, the values of λ_0 and λ_1 must be restricted in order to identify the model parameters of interest¹⁸. Restricting $\lambda_0 = 0$ and $\lambda_1 = 1$ yields (33).

Now that the mapping from latent skills to observed data is complete, I can discuss the simulation process for child cognitive skills. The initial period for each household takes place in 1997 when their child is some initial age t_0 . I assume that the household has an initial prior for $p(k_t)$ that is drawn from a Beta(1,1) distribution and I use $k_{t_0}^*$ as the initial condition for each household. From this initial measure, I update the household's priors for p according to $\tilde{p} \sim \text{Beta}(1 + k_t^*, 57 - k_t^* + 1)$. The initial latent continuous measures for child skills are then given by

$$k_{t_0} = \frac{\tilde{p}}{1 - \tilde{p}} \quad (35)$$

From these initial values for k_t , I can construct a sample path for each household, and simulate decisions and outcomes according to the model structure laid out in Section 2 from the child's initial age, t_0 , up through when the child is 18 years old.

4.6 Measurement of Child Non-Cognitive Skills

Children's non-cognitive skills are represented by their externalizing and internalizing components of the Behavior Problem Index. Each wave of the CDS, the Primary Caregiver,

¹⁸See [Agostinelli and Wiswall \(2025\)](#) for a detailed explanation in the context of a model of child skill development.

is asked a series of questions regarding the behavior of their child. Internalizing behaviors refer to behaviors such as being withdrawn, sad, or anxious, where externalizing behaviors refer to aggressiveness and impulsivity. I reorder these scales so that higher scores refer to better behaved children, and divide them into the top 50 and bottom 50th percentiles in each wave, to correspond to high and low skills. I use the 1997 measure available in the CDS as initial conditions for the child, and then simulate non-cognitive skills according to the model structure discussed above.

5 Identification

5.1 Cognitive Skill Production Parameters

The parameters governing the production of child cognitive skills are allowed to vary with the age of the child and, for some parameters, the level of education of the parents. Identification of the skill production parameters would be straightforward if I observed all inputs of the production function and child outcomes at both age t and age $t+1$. Instead, when a child is age t in 1997 (or 2002), I observe parent and child time investment, the parents' choice of a parenting style, and the measure of child skills in that year, and I do not observe parental monetary investments.¹⁹ Additionally, I do not observe child skill measures in consecutive years, instead they are observed every five years.

There is substantial variation in parent and child time investment decisions, parenting style decisions, and child test scores across households and across childhood that allows for identification of the structural parameters governing the cognitive skill production function. Given that parental preferences over their leisure time, consumption, and child skills are fixed throughout childhood, and uncorrelated with household observables (outside of preferences over parenting styles), one can use variation in parental time investment across child age and by level of parental education with variation in test score outcomes to identify the parameters governing the marginal product of parental time investment. Similarly, there is variation in child study time across children and across ages, and this variation coupled with variation in test scores pins down the marginal product of child study time. Variation in child test scores at age t and age $t+5$ also aids in identification of the marginal product of the stock of child skills.

Total factor productivity is allowed to vary with child age and takes a logistic form. TFP is additively separable and does not enter the first order conditions for the parents or the child. This means it is exogenous to household choices regarding time and monetary

¹⁹While the CDS does have some measures of monetary investments, they are quite noisy and typically not used as moment conditions in the literature.

investments. Changes in TFP would influence levels of child cognitive skills generated by the model, and therefore, the average score on the LW exam at every child age allows for identification of the parameters governing the TFP process.

All that remains to identify in the child skill production process is the marginal product of child goods investment. While I do not have explicit data on the monetary expenditures parents make in their children, I reference data from the Consumer Expenditure Survey and match simulated moments from those in the survey.²⁰ Theoretically, the marginal product of monetary investments in children can be pinned down through remaining variation in child test scores that is not explained by parental or child choices, or TFP.

5.2 Non-Cognitive Skill Production

Binary measures for child non-cognitive skills are available in five year intervals. The functional form for the transition probabilities for non-cognitive skill N are

$$p_{t+1}^N = \frac{\exp(\beta_0^N + \beta_1^N t + \beta_2^N (ps_t = AR) + \beta_3^N (ps_t = AT) + \beta_4^N B_t^e + \beta_5^N B_t^n)}{1 + \exp(\beta_0^N + \beta_1^N t + \beta_2^N (ps_t = AR) + \beta_3^N (ps_t = AT) + \beta_4^N B_t^e + \beta_5^N B_t^n)}$$

for $N \in \{e, n\}$. There is variation in measures of child non-cognitive skills, child age, lagged child non-cognitive skills, as well as the parents' choice of parenting style. The identification issue is similar to the one for cognitive skills, that measures are not observed in consecutive years. If they were, a logistic regression of non-cognitive skills on lagged parenting style decisions and skills would produce consistent estimates for the structural parameters. Variation in the data regarding non-cognitive skills, parenting style choices, and the age of children still allows for these parameters to be identified, and in practice unconditional transitions for each dimension of non-cognitive skills, as well as transitions conditional on the choice of parenting style and lagged skills are targeted in estimation.

5.3 Wages and Non-Labor Income

Parent log-wage offers are drawn from a Multivariate Normal distribution:

$$\begin{pmatrix} \ln(w_t^f) \\ \ln(w_t^m) \end{pmatrix} \sim Normal \left(\begin{pmatrix} \eta_t^f \\ \eta_t^m \end{pmatrix}, \begin{pmatrix} \sigma_f^2 & \rho\sigma^w \\ \rho\sigma^w & \sigma_m^2 \end{pmatrix} \right)$$

where

$$\eta_t^j = \eta_0^j + \eta_1^j A_t^j + \eta_3^j E^j$$

²⁰This follows from [Del Boca et al. \(2026\)](#). I also match moments concerning the correlation in accepted hourly wages and child observed test scores.

for $j \in \{m, f\}$, A_t^j represents parent j 's age when the child is t years old, and E^j is parent j 's years of schooling. In the main PSID files, I have access to labor supply decisions, accepted wages, as well as information about the age of parents and their years of schooling. The main identification issue is the same that plagues most researchers in labor economics, I do not observe wage offers for individuals who choose not to work. The issue of identification is slightly ameliorated because I assume preferences over leisure, consumption, and child skills are common across parents and do not allow work experience to influence the wage process and ²¹. Using the data generating process outlined above, I am able to identify the parameters governing the wage offer distribution by simultaneously estimating them with the remaining parameters in the model. I observe variation in accepted wages by parent age and education, as well as variation in accepted wage offers within households over time, simulating both the wage offer distribution as well as parent labor supply decisions allows for identification of said process. The key moments used in identification are the mean and standard deviation of parent (accepted) wages, average wages within age bins, average wages within parent education bins, correlations across parent wages, and the standard deviation of each parent's wage.

The non-labor income process is simpler to identify as there are no dynamic processes involved and no selection issues to deal with. The non-labor income process is estimated in a first stage before estimating the remaining model parameters. Non-labor income is observed for all households in which it appears in the PSID (biannually after 1997 corresponding to the previous year, that is, an observation in 1999 corresponds to non-labor income in 1998), therefore, estimation of the two stage non-labor income process is straightforward. Given that there is variation in household non-labor income by father's age and education, a logistic regression for positive non-labor income on father's age, education, and age-squared yields consistent estimates of the first stage parameters of the non-labor income process. Then, conditional on a household having positive non-labor income, I again use the variation in the levels of non-labor income by the father's characteristics to identify the parameters that govern the level of non-labor income a household has in a given period.

5.4 Preferences

Parents have preferences over their leisure time (α_t^m and α_t^f), consumption (α_c), their child's skills (α_k), and derive utility cost from their choice of parenting style ($\xi(ps_t, \tau_t^c)$). Children have preferences for leisure time $\omega(B_t^e, B_t^n)$ and their skills ($1 - \omega(B_t^e, B_t^n)$) which are positive, sum to one, and are allowed to depend on the stock of child non-cognitive skills.

²¹See [Del Boca et al. \(2014\)](#), [Del Boca et al. \(2026\)](#), and [Brilli \(2022\)](#) who make the same assumption with similar samples of the PSID-CDS data.

I begin with the identification of the children's preference for leisure. Their choice of study time at any age t when they face a permissive parenting style is

$$\tau_t^c(p s_t = P) = \frac{\beta^c D_{t+1}^c \delta_t^c}{\omega(B_t^n, B_t^e) + \beta^c D_{t+1}^c \delta_t^c} (112 - \tau_t^f - \tau_t^m - s_t)$$

Inverting this equation and solving for $\omega(B_t^e, B_t^n)$ yields

$$\omega(B_t^e, B_t^n) = \left(\frac{(112 - \tau_t^f - \tau_t^m - s_t)}{\tau_t^c(p s_t = P)} - 1 \right) \beta^c D_{t+1}^c \delta_t^c$$

or

$$\frac{\exp(\omega_0 + \omega_e B_t^e + \omega_n B_t^n + \omega_{en} B_t^e B_t^n)}{1 + \exp(\omega_0 + \omega_e B_t^e + \omega_n B_t^n + \omega_{en} B_t^e B_t^n)} = \left(\frac{(112 - \tau_t^f - \tau_t^m - s_t)}{\tau_t^c(p s_t = P)} - 1 \right) \beta^c D_{t+1}^c \delta_t^c$$

Therefore, variation in child study time conditional on internalizing and externalizing non-cognitive skills allows for identification of children's preference for leisure, and skills, as the parameters must sum to one.

The remaining parameters to identify are the ones that govern parent preferences for leisure, consumption, and their child's skills, $\alpha = \{\alpha_l^m, \alpha_l^f, \alpha_c, \alpha_k\}$, which are common to all parents, and the costs for adopting an authoritarian or an authoritative parenting style $\xi(p s_t, \tau_t^c)$. First, consider a household that always adopts the same parenting style in each period. While this does not yield information about $\xi(p s_t, \tau_t^c)$, the variation in labor supply and time investment decisions aid in the identification of the parameters governing $\alpha = \{\alpha_l^m, \alpha_l^f, \alpha_c, \alpha_k\}$. Conditional on the choice of parenting style, the numerical solution for (τ_t^m, τ_t^f) pins down labor supply. Since these parameters are common across households, I can use information about average parental time investment, average labor supply, and the fraction of parents supplying any labor to identify them.

Identification of the parameters governing $\xi(p s_t, \tau_t^c)$ requires more care. The marginal cost of authoritarian and authoritative parenting is:

$$\xi(AR) = \exp(\kappa^i)$$

$$\xi(AT) = \exp(\kappa^i + \kappa^w)$$

where

$$\kappa^i = \mu^i + \epsilon^i$$

and

$$\epsilon^i \sim N(0, \sigma^i)$$

and $\kappa^w > 0$, the cost of warm monitoring, is common to all parents. Consider the solution for the parents' choice of child study time under an authoritarian parenting style:

$$\tau_t^c = \frac{\beta^p D_{t+1}^p \delta_t^c - (1 - \varphi) \xi(AR)}{\varphi \omega(B_t^n, B_t^e) + \beta^p D_{t+1}^p \delta_t^c - (1 - \varphi) \xi(AR)} \tilde{T}$$

Solving for $\xi(AR)$ yields

$$\xi(AR) = \frac{\beta^p D_{t+1}^p \delta_t^c}{(1 - \varphi)} - \frac{\varphi \omega(B_t^e, B_t^n)}{\frac{\tilde{T}}{\tau_t^c} - 1} \quad (36)$$

or for household i , the cost of authoritarian parenting is

$$\exp(\kappa^i) = \frac{\beta^p D_{t+1}^p \delta_t^c}{(1 - \varphi)} - \frac{\varphi \omega(B_t^e, B_t^n)}{\frac{\tilde{T}}{\tau_t^c} - 1} \quad (37)$$

and authoritative parenting is

$$\exp(\kappa^i + \kappa^w) = \frac{\beta^p D_{t+1}^p \delta_t^c}{(1 - \varphi)} - \frac{\varphi \omega(B_t^e, B_t^n)}{\frac{\tilde{T}}{\tau_t^c} - 1} \quad (38)$$

Therefore, (μ^i, σ^i) are pinned down by the mean and variance of child study time conditional on parents choosing an authoritarian parenting style. Since intensive parenting is drawn from a single distribution, the extra cost of warmth associated with authoritative parenting is identified by average study time conditional on authoritative parenting.

6 Estimation

I estimate the structural parameters of the model outlined above using Simulated Method of Moments (SMM). The estimation routine is done in two stages, first, recall that the non-labor income process only depends on the father's demographic characteristics, and is exogenous with respect to household decisions, this allows me to estimate the parameters governing the non-labor income process separately from the remaining model parameters in a first stage, and then use these results in estimation of the full structural model. Let P be the number of parameters to be estimated in the second stage, and Ω be a P dimensional vector of structural parameters. My estimator $\hat{\Omega}_{\text{SMM}}$ is defined as

$$\hat{\Omega}_{\text{SMM}} = \text{argmin}_{\Omega} (m(d) - m(\Omega))' W (m(d) - m(\Omega)) \quad (39)$$

where $m(d)$ is a $Z > P$ dimensional vector of sample moment conditions taken from the data, and $m(\Omega)$ is a corresponding set of moment conditions generated by a simulated data set at the parameter vector Ω . The weighting matrix, W , is a diagonal matrix with weights that are bootstrapped standard deviations of the sample moments²².

I use 30 moments to estimate 9 parameters that govern the non-labor income process. I match moments concerning the probability that a household has non-labor income in a given period as a function of father’s characteristics for the logit step, and moments that give information about the mean and standard deviation of the level of non-labor income a household has as a function of the father’s characteristics. The second stage of the model has 51 parameters to be estimated and I use 255 moment conditions in my estimation routine. Some of the moments I match are the mean and standard deviation of time allocation decisions for the father, mother, and child, study time by parenting style, the frequency of parents supplying labor as well as the mean and standard deviation of hours worked, and average skills by child age. Additionally, I match moments about parenting style choices, such as the frequency each style is adopted conditional on child age and the education level of the mother.²³

Once the parameters governing the non-labor income process have been estimated, estimation of the full structural model is straightforward. I simply take a guess of the parameter vector Ω , and given this guess, simulate the decisions of each household according to the structural model explained in Section 2. I simulate each household R times, where R is set equal to 5, and the only difference for each copy $r \in \{1, \dots, R\}$, are the error draws for the wage, non-labor income, and measurement error processes. With this simulated data in hand, I compute the moment conditions and evaluate the weighted distance according to (39). This process then iterates until a convergence criterion is satisfied²⁴. To compute the standard errors of my estimates, I re-estimate the model 50 times using non-parametric bootstrapped samples with replacement and compute standard errors following Train (2009) (p. 201).

7 Results

A discussion of the estimates pertaining to preferences and skill production are presented here. A discussion of the results for the wage and non-labor income processes have been left for the Appendix.

²²I use 200 bootstrap samples to compute the weight matrix.

²³The full list of the moment conditions used are shown in Appendix E.

²⁴I use the NLOpt NelderMead algorithm on Julia v1.12.1 to estimate the structural model. The convergence criterion is set to $1e-5$.

7.1 Preferences

Tables 5 and (6) below display the estimated parameters for parent and child preferences. Recall that parental altruism (φ) is exogenously set such that $\varphi = 1/3$. Similar to others in the literature, I find that parents place the highest weight on father’s leisure (0.338), followed by parental consumption (0.326), and child cognitive skills receive the lowest weight (0.095). While these parameters are common across households, I allow for heterogeneity in the cost of intensive parenting. While difficult to interpret the untransformed parameters, the median marginal cost of increasing child study time while adopting an authoritarian parenting style across simulations is about 0.0001. A high cost for authoritarian parenting could either indicate that the parents face some sort of constraint that makes it difficult to adopt an authoritarian parenting style, or it could capture parents who are quite averse to an intensive parenting style. Authoritative parenting comes with an additional cost that varies with the level of education of the mother, and is constrained to be positive, implying that there is an additional cost to adopting an intensive parenting style that is also warm toward the child. This cost is about 1.84 times higher for households where the mother has at most a high school degree. Where this higher cost is derived from, the model cannot speak to, it could be that higher educated parents have taken parenting classes that make them more skilled at adopting an authoritative parenting style. It could be that elements of the communities or school districts where higher educated parents live are more suitable for an authoritative parenting style (or make the cost of shifting away from authoritarian parenting greater, as indicated by (Harris (2024))).

Table 5. **Parent Preferences**

	Parameter	Estimate	(Std Error)
	Mother’s Leisure (α_l^m)	0.242	(0.002)
	Father’s Leisure (α_l^f)	0.338	(0.002)
	Parent Consumption (α_c)	0.326	(0.004)
	Parent Preference Cognitive Skill (α_k)	0.095	(0.001)
	Parent Terminal Value (ψ^p)	0.840	(0.033)
	Mean Intensive Cost (μ_i)	-8.902	(0.386)
	SD Intensive Cost (σ^i)	5.606	(0.442)
	Warmth Cost Some College (c_h^w)	1.631	(0.497)
	Warmth Cost HS (c_l^w)	2.854	(0.330)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

Children’s preferences are endogenously determined and can take on one of four values in any given period. These values, along with the child’s terminal value on their cognitive skills,

are presented in Table 6. The results show that the stock of children’s non-cognitive skills have a significant effect on their preference for leisure, as large differences are observed when comparing children who have high or low skills along both dimensions. An interesting result is that children who have low externalizing and internalizing skills actually place less weight on their leisure time (therefore, more weight on their own cognitive skills) than children who also have low externalizing skills, but high internalizing skills (0.894 compared to 0.975). Parents who report a child that has high internalizing skills will (in their own view as this measure is subjective) see a child who they would not describe as anxious. It is possible that children who are anxious may care more about their human capital as their future skill development may be a source of their anxiety. Children who have high externalizing skills place a lower weight on their leisure time. These values are as low as 0.206 when they also have low internalizing skills, and 0.732 when they have high non-cognitive skills along both dimensions.

Table 6. **Child Preferences**

	Parameter	Estimate	(Std Error)
	Low E Low N (ω_0)	0.894	(0.014)
	High E Low N ($\omega_0 + \omega_e$)	0.206	(0.010)
	Low E High N ($\omega_0 + \omega_n$)	0.975	(0.011)
	High E High N ($\omega_0 + \omega_e + \omega_n + \omega_{en}$)	0.732	(0.024)
	Terminal Value (ψ^c)	1.603	(0.195)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

7.2 Skill Production

Table 7 presents the parameter estimates for the child’s cognitive skill production function. While awkward to interpret, the model is able to generate patterns observed in the literature. The marginal product of parental time investment is decreasing with age (captured by the negative slope coefficients), while children’s own time investment becomes more productive. Additionally, the more educated parents are, the more productive their time investment is. Also consistent with the literature is the result that the mother’s marginal product of time investment is higher than the father’s throughout childhood, however, the gap decreases with child age.

This paper is perhaps the first to allow for parental education to affect the child’s marginal product of study time, and I find that the mother’s education is positively associated with child study time productivity. This could be for a multitude of reasons, it could be that children of higher educated mothers simply have a better natural ability to study or they

might have access to better schools/peers and therefore have developed better study habits, it could be that they have access to higher quality materials at home, or that they have a better latent ability to study that is derived from their higher educated mother. Lagged skills and child goods both have a marginal product that is increasing with the age of the child. The result that the marginal product of lagged skills is increasing in age supports the result in [Cunha et al. \(2010\)](#) that early investments in skills matter because it will become harder for children to catch up in later years

Table 7. **Cognitive Skill Production Parameters**

	Parameter	Estimate	(Std Error)
Father (δ_t^f)	Intercept	-1.617	(0.029)
	Slope (Age)	-0.019	(0.002)
	Education	0.014	(0.001)
Mother (δ_t^m)	Intercept	-0.390	(0.010)
	Slope (Age)	-0.059	(0.001)
	Education	0.003	(0.000)
Child (δ_t^c)	Intercept	-6.908	(0.272)
	Slope (Age)	0.228	(0.020)
	Education	0.042	(0.001)
Child Goods (δ_t^e)	Intercept	-2.511	(0.003)
	Slope	0.058	(0.001)
TFP (δ_t^R)		0.023	(0.009)
		0.146	(0.002)
		4.241	(0.412)
		4.288	(0.065)
Lagged Skill (δ_t^k)	Intercept	-0.234	(0.002)
	Slope	0.005	(0.000)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

7.3 Non-Cognitive Skills

Table 8 presents the estimates for the parameters that govern the production of internalizing non-cognitive skills. Already having high internalizing skills increases the probability that internalizing skills will be high next period, though the opposite is true for high externalizing skills. It might be that well-behaved children (high externalizing skills) are more likely to be anxious (low-internalizing skills), which could be driving this result. Finally, authoritarian

parenting is the “best” parenting style in terms of increasing the probability of having high internalizing skills, however, the relative increase in the probabilities are small, relative to permissive parenting, authoritarian and authoritative parenting do not have much influence over internalizing non-cognitive skills.

Table 8. **Internalizing Skill Production**

Parameter	Estimate	(Std Error)
Intercept (p_0^n)	0.822	(0.038)
Age (p_1^n)	-0.036	(0.001)
AR (p_2^n)	0.001	(0.000)
AT (p_3^n)	0.000	(0.000)
Lagged N (p_4^n)	0.229	(0.108)
Lagged E (p_5^n)	-0.142	(0.011)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

Parenting style has a greater influence over externalizing non-cognitive skills as show in Table 9. Consistent with the literature, authoritarian parenting increases the probability that a child has low externalizing skills, meaning it is more likely to cause children to have behavior issues such as acting out. Interestingly, authoritative parenting has an even more negative coefficient, an unexpected result. A reason could be that the measures for non-cognitive skills are subjective and therefore I am observing the parents’ perception of child non-cognitive skills. In both cases, the absence of an interaction term may be driving the unexpected results, as it could be the case that intensive parenting styles are beneficial or harmful for children at different levels of non-cognitive skill, and I am only reporting the dominant average effect.

Table 9. **Externalizing Skill Production**

Parameter	Estimate	(Std Error)
Intercept (p_0^e)	2.187	(0.095)
Age (p_1^e)	-0.120	(0.002)
AR (p_2^e)	-0.133	(0.001)
AT (p_3^e)	-0.138	(0.001)
Lagged E (p_4^e)	-0.531	(0.091)
Lagged N (p_5^e)	2.056	(0.226)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

In both cases, age is correlated with lower non-cognitive skills, and this effect is stronger for externalizing skills. Further, lagged externalizing skills have a negative effect on future

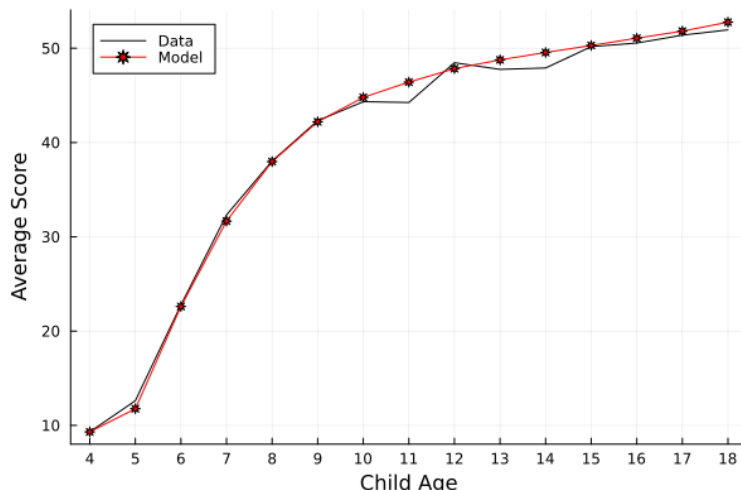
externalizing skills, but internalizing skills have a large positive influence. These results again may be driven by the fact that the measures for non-cognitive skills are subjective, and, internalizing behaviors could be more difficult for parents to observe.

As discussed in the introduction, this paper presents a partial equilibrium model of child human capital formation, and care should be taken when interpreting the results in this section. Parenting style decisions likely also respond to environmental effects that are outside of the home, which are unobserved in this framework, therefore, these parenting style effects may be picking up other environmental effects that are absent and should be considered in future work.

8 Model Fit

8.1 Child Skills

Figure 3. Fit of Average Child Skill Evolution



Notes: Data refers to average child LW scores from the author's CDS sample. Model refers to the average fit of simulated child scores generated by the structural parameters estimates.

Figure 3 above shows that the model is able to accurately capture the upward sloping and concave nature of the evolution of children's cognitive skills. This suggests that while a simple log-linear production function has been implemented, it is still able to capture the trends present in the data.²⁵

Table 10 shows that the model fits the data in terms of average children's non-cognitive skills, with externalizing non-cognitive skills having a closer fit than internalizing non-

²⁵Notice that while the model begins when children are 4 years old, this data point is omitted from the figure. This is because the first test score is used as an initial condition, so they model perfectly matches the data.

cognitive skills. A full list of moments concerning the fit of non-cognitive skills can be found in Appendix E, but the model is able to capture average behavior problems conditional on lagged parenting style, as well as the stock of lagged non-cognitive skills.

Table 10. **Fit of Average Child Non-Cognitive Skills**

	Skill	Data	Model
	Avg Externalizing	0.66	0.68
	Avg Internalizing	0.62	0.56
	Avg BPIE if Age Less than 12	0.63	0.64
	Avg BPIN if Age Less than 12	0.61	0.50
	Avg BPIE Age 12 and up	0.69	0.70
	Avg BPIN Age 12 and Up	0.63	0.60

Notes: Data refers to average child BPI measures from the author's CDS sample. Model refers to the average fit of simulated child non-cognitive skills generated by the structural parameters estimates.

8.2 Time Investment

Table 11 displays how the model performs in terms of average time investment by parents and their children. The model captures the decreasing time investment of parents as children age, as well as the increasing self-investment by children.

Table 11. **Fit of Average Household Time Investment**

	Age	Data	Model
Mom	4-7	23.43	21.97
	8-11	16.15	18.87
	12-14	17.07	16.48
	15-18	11.96	13.25
Dad	4-7	9.18	7.98
	8-11	8.39	7.90
	12-14	6.73	7.85
	15-18	6.32	7.29
Child	4-7	0.83	1.11
	8-11	3.25	2.69
	12-14	5.30	4.76
	15-18	6.36	6.26

Notes: Data refers to average household time investment from the author's CDS sample. Model refers to the average fit of simulated time investment generated by the structural parameter estimates.

Table 12 shows average child study time conditional on age and parenting style. The

model is able to replicate the conditional study time patterns, both in the aggregate, and across the two age bins. In fact, in the older age bin, the model is able to produce the correct ordering, with study time being the highest for children facing an authoritative parenting style, and lowest for those facing an authoritarian parenting style. While this is not the case in the other two frames, the moments are closely matched. The reason for the deviation could be that the costs to adopt authoritarian and authoritative parenting are fixed, and do not vary with the age of the child, as one could believe that certain parenting styles become more difficult or perhaps easier to implement as children age²⁶.

Table 12. **Fit of Average Child Time Investment by Parenting Style**

Age	Parenting Style	Data	Model
All	Authoritarian	4.37	4.32
	Authoritative	4.37	4.20
	Permissive	4.48	4.47
4-11	Authoritarian	2.25	1.93
	Authoritative	2.44	2.18
	Permissive	2.36	2.55
12-18	Authoritarian	5.86	5.25
	Authoritative	6.24	5.92
	Permissive	6.04	5.59

Notes: Data refers to average household time investment from the author's CDS sample. Model refers to the average fit of simulated time investment generated by the structural parameter estimates.

Parenting Style

Table 13 shows that the model is able to generate the parenting style patterns observed in the data, both conditional and unconditional on the age of children. Unconditional on child age, the model performs exceptionally, generating the result that an authoritative parenting style is adopted most often, with permissive and authoritarian parenting being adopted with about equal frequencies. When conditioning on age the model still performs well, authoritative parenting is adopted slightly less often than the model predicts when children are 11 and younger, and slightly more often when they are 12 and older, and the reverse trend is observed for authoritarian parenting. This is again likely driven by the assumption that the parenting style costs are not allowed to vary with the age of the child.

²⁶There could also be switching costs that are not present in this framework.

Table 13. **Parenting Style Fit**

Age	Parenting Style	Data	Model
All	Authoritarian	0.27	0.27
	Authoritative	0.46	0.45
	Permissive	0.27	0.28
4-11	Authoritarian	0.24	0.20
	Authoritative	0.50	0.54
	Permissive	0.25	0.26
12-18	Authoritarian	0.29	0.32
	Authoritative	0.42	0.40
	Permissive	0.29	0.28

Notes: Data refers to average household time investment from the author's CDS sample. Model refers to the average fit of simulated time investment generated by the structural parameter estimates.

9 Counterfactual: Free After School Homework Program

Consider an after school homework program that exogenously sets a lower bound on child study time without parental involvement. This changes the choice problem that households face. Total child study time is now defined as

$$\tilde{\tau}_t^c = \tau_t^{c,HW} + \tau_t^c$$

where $\tau_t^{c,HW}$ refers to the time the child spends in the homework program and $\tilde{\tau}_t^c$ is now total child study time. The production function is now given by

$$\ln(k_{t+1}) = \ln(R_t) + \delta_t^m \ln(\tau_t^m) + \delta_t^f \ln(\tau_t^f) + \delta_t^c \ln(\tilde{\tau}_t^c) + \delta_t^g \ln(g_t) + \delta_t^k \ln(k_t)$$

Taken together, the solution to the child's problem for their own study time conditional on their parents choosing a permissive parenting style is

$$\tau_t^c = \frac{\tilde{T}T FV_{t+1}^c - \omega(B_t^e, B_t^n) \tau_t^{c,HW}}{\omega(B_t^e, B_t^n) + FV_{t+1}^c} \quad (40)$$

where $\tilde{T}T = 112 - \tau_t^m - \tau_t^f - s_t - \tau_t^{c,HW}$ and $FV_{t+1}^c = \beta^c D_{t+1}^c \delta_t^c$. As a result of the homework policy, a child may choose to spend zero hours per week at home in educational activities if conditions are such that the numerator of (40) is weakly negative.

Of course, this changes the parents' decision problem as well. At no cost, their child's

study time has increased, therefore, they may not need to spend the effort to adopt an intensive parenting style to increase study time. A potential trade-off could be that adopting a different parenting style changes the trajectory of their future non-cognitive skills, which will impact future study time and cognitive skill accumulation.

Table 14 presents summary statistics for the after school policy. There are three main takeaways, first, the average terminal stock of children’s cognitive skills increases, however, consistent with the result in Eren and Henderson (2008), not all children see an improvement in their cognitive skills as a result of the increased study time (42% see an increase in their final cognitive skills and about 5% see a reduction in the final stock of their cognitive skills)²⁷. The average final stock of children’s non-cognitive skills remains mostly unchanged, with only about 1% of children seeing an increase or decrease in their externalizing skills, and internalizing skills are even stickier. This is likely driven by the result that parenting style decisions have a greater influence on externalizing behaviors, and as one can see, this policy does augment parenting style decisions. With the burden of having to adopt an intensive parenting style to increase study time, parents shift away from authoritarian and authoritative parenting, opting for a permissive parenting style²⁸.

Table 14. After School Homework Policy

	Baseline	Homework Policy
Average Terminal (Log) Cognitive Skills	2.92	3.01
Avg Terminal Externalizing Skills	0.62	0.62
Avg Terminal Internalizing Skills	0.59	0.59
Average Child Time Investment	4.31	5.62
Fraction Cognitive Skills Improved	NA	0.42
Fraction Cognitive Skills Reduced	NA	0.05
Fraction Externalizing Improved	NA	0.01
Fraction Externalizing Reduced	NA	0.01
Fraction Internalizing Improved	NA	0.00
Fraction Internalizing Reduced	NA	0.00
Fraction Authoritarian	0.27	0.10
Fraction Authoritative	0.45	0.28
Fraction Permissive	0.28	0.62

Notes: Baseline refers to the simulated data generated from the model using the structural parameters estimated. Homework Policy refers to a policy where I artificially increase child study time without the need for parental intervention.

²⁷In simulation, I round cognitive skills to their nearest grid point, otherwise, there would likely be a larger fraction of households that see a change in the final stock of their child’s cognitive skills.

²⁸This of course, hinges upon the parenting style classification laid out in Section 3, if parenting styles are mis-classified, this would influence both the parameters governing non-cognitive skill production and the monitoring cost parameters. This would alter both household time allocation decisions and child skills.

10 Conclusion

This paper extends the economics literature on child skill development by allowing parents to choose one of the three major parenting styles defined by [Baumrind \(1966\)](#), namely, authoritarian, authoritative, and permissive. In this framework, authoritarian and authoritative parents acts to constrain the child’s study time decision, and all three parenting styles are allowed to affect the child’s non-cognitive skill development. Children’s preferences for leisure are anchored to their non-cognitive skills, therefore, parenting style decisions are allowed to mold the preferences of their children, influencing both future child and parenting decisions.

Simulated data generated by the structural parameters fit the patterns observed in the data. A novel feature of this work was linking children’s preferences to their non-cognitive skills, which are anchored to the Behavior Problem Index. This allowed for parenting style decisions to have lasting effects on their children, as they may adopt an intensive parenting style until their child’s preferences are more in line with what their parents desire. Using the parameter estimates, I simulate a counterfactual after school homework policy that exogenously increases child study time. Parents respond to this increase in study time by opting for permissive parenting, as they no longer need to exert effort to constrain their child’s choice. The policy increases the average terminal stock of children’s cognitive skills, without having as great an effect on their non-cognitive skills.

The model is not without limitations, while I make the distinction that parents take on an extra cost to adopt an authoritative parenting style to engage in “warm” monitoring of their child, this warmth does not show up in the current period. While there could be long-term benefits to warm parenting, they are likely to affect the child in the current period, and, if parents are altruistic, they would see benefits as well. Additionally, it may prove fruitful to analyze certain components of the Behavior Problem Index. While it is interesting to look at both internalizing and externalizing behaviors, the Behavior Problem Index provides information about specific behaviors that may be of interest to both economists and non-economists. Further, this model presents a partial equilibrium framework of child cognitive and non-cognitive skill development, future research should expand upon this work to include the role of factors outside of the home such as the school and environment that households live in.

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Appendix

A Household Equilibrium

Decision rules for all members of the household are solved via backward induction. The child, who is unaware that their preferences could change with time, solve their problem as those in [Del Boca et al. \(2014\)](#). Parents, who are aware that their child's preferences are endogenous, the backward recursion process requires more care, and follows that in [Del Boca et al. \(2026\)](#). How the household's problem is solved each period will depend on whether or not the parents choose to adopt an intensive parenting style or not. If parents are permissive, their child choosing study time on their own, if their parents choose an authoritarian or authoritative parenting style, the parents choose the child's study time for them.

A.1 Period T Problem

In the final period, the child solves

$$V_T^c(\Gamma_T | a_T^p) = \max_{\tau_T^c} u(l_T^c, B_T^e, B_T^n) + \beta^c E_T V_{T+1}^c(\tilde{\Gamma}_{T+1} | \tau_T^c, a_T^p, \Gamma_T)$$

subject to their time constraint. Given that the child does not believe their preferences will change, their solution if they face a permissive parenting style is:

$$\tau_T^c = \frac{\beta^c D_{T+1}^c \delta_T^c}{\omega(B_T^e, B_T^n) + \beta^c D_{T+1}^c \delta_T^c} \tilde{T}_T$$

where

$$D_{T+1}^c = \frac{\partial V_{T+1}^c}{\partial \ln k_{T+1}} = \psi^c(1 - \omega(B_T^e, B_T^n))$$

given that children assume their preferences will remain unchanged, I can compute the following sequence via backward induction:

$$D_{T+1}^c = \frac{\partial V_{T+1}^c}{\partial \ln k_{T+1}} = \psi^c(1 - \omega(B_T^e, B_T^n))$$

$$D_T^c = (1 - \omega(B_T^e, B_T^n)) + \beta^c D_{T+1}^c \delta_T^k$$

and for any period t:

$$D_t^c = (1 - \omega(B_t^e, B_T^n)) + \beta^c D_{t+1}^c \delta_t^k$$

I am able to compute this sequence for all for combinations of (B_t^e, B_T^n) .

The parents' problem requires more care, as they are aware that their child's preferences

are endogenous. In the final period, they choose $a_T^p = \{h_T^f, h_T^m, \tau_T^m, \tau_T^f, e_t, ps_T\}$ to solve:

$$V_T^c(\Gamma_T) = \max_{a_T^p | \tau_T^c(a_T^p)} (1-\varphi)u^p(l_T^m, l_T^f, c_t, k_T) + \varphi u^c(l_T^c, k_T, B_T^e, B_T^n) + \beta^p E_{T+1} V_{T+1}^p(\Gamma_{T+1} | a_T^p, \tau_T^c(a_T^p), \Gamma_T)$$

where

$$E_T V_{T+1}^p(\Gamma_{T+1} | a_T^p, \tau_T^c(a_T^p), \Gamma_T) = \psi^p E_T(\alpha_k + (1 - \omega(B_{T+1}^e, B_{T+1}^n))) \ln k_{T+1}$$

as the parents care both about the cognitive skill of their child as well as how much their child values their cognitive skills. The expectation is taken over the stocks of child non-cognitive skills. Unlike [Del Boca et al. \(2014\)](#), I cannot derive analytical solutions for all of the parents' choices, however, conditional on numerical solutions for parent time investment, all other choices outside of the discrete choice of parenting style have an analytical solution.

Suppose the parents choose a permissive parenting style, conditional on (τ_T^m, τ_T^f) , the remaining parental choices are:

$$h_T^f(\tau_T^m, \tau_T^f) = \frac{w_T^f(112 - \tau_T^f)((1 - \varphi)(\alpha_l^m + \alpha_c) + \beta^p D_{T+1}^p \delta_T^e) - (1 - \varphi)\alpha_l^f(w_T^m(112 - \tau_T^m) + y_T)}{w_T^f((1 - \varphi)(\alpha_l^f + \alpha_l^m + \alpha_c) + \beta^p D_{T+1}^p \delta_T^e)}$$

and

$$h_T^m(\tau_T^m, \tau_T^f) = \frac{w_T^m(112 - \tau_T^m)((1 - \varphi)(\alpha_l^f + \alpha_c) + \beta^p D_{T+1}^p \delta_T^e) - (1 - \varphi)\alpha_l^m(w_T^f(112 - \tau_T^f) + y_T)}{w_T^m((1 - \varphi)(\alpha_l^f + \alpha_l^m + \alpha_c) + \beta^p D_{T+1}^p \delta_T^e)}$$

where the observed choice is

$$h_T^{j,*}(\tau_T^m, \tau_T^f) = \max\{h_T^j(\tau_T^m, \tau_T^f), 0\}$$

With these choices in hand (that are both constrained to be non-negative), the solution for child goods, is a fraction of total household income

$$e_T(\tau_T^m, \tau_T^f) = \frac{\beta^p D_{T+1}^p \delta_T^e}{(1 - \varphi)\alpha_c + \beta^p D_{T+1}^p \delta_T^e} (w_T^f h_T^{f,*}(\tau_T^m, \tau_T^f) + w_T^m h_T^{m,*}(\tau_T^m, \tau_T^f) + y_T)$$

where

$$D_{T+1}^p = \frac{\partial(V_{T+1}^p)}{\partial \ln k_{T+1}} = \alpha_k + (1 - \omega(B_{T+1}^e, B_{T+1}^n))$$

If the parents choose an authoritarian or authoritative parenting style, they constrain

the choice of their child by choosing their study time:

$$\tau_T^c = \frac{\beta^p D_{T+1}^p \delta_T^c - (1 - \varphi) \xi^p}{\varphi \omega(B_T^e, B_T^n) + \beta^p D_{T+1}^p \delta_T^c - (1 - \varphi) \xi^p} \tilde{T}$$

A.2 Backward Induction

To be able to compute solutions in a similar manner in other periods, I use a numerical approximation for the parents' value functions following [Del Boca et al. \(2026\)](#). In the terminal period, the parents will choose $a_T^{p,*}$, the utility maximizing set of parenting style, time, and monetary investment decisions, and this will produce a corresponding value $V_T^{p,*}(\Gamma_T)$. In order to keep the model tractable, I approximate their value function. Each non-cognitive skill has two potential outcomes, or grid points, and I define a grid of K points for children's cognitive human capital²⁹. In total, there are Kx2x2 grid points, however, the functional form of the model implies that the parents' value function is separable in future log cognitive skills, reducing the computational burden of the model³⁰.

The solution iterates as follows:

1. Loop over the child's stock of externalizing and internalizing non-cognitive skills
2. Loop over the three parenting style choices
3. At this point, the following can be calculated

$$\begin{aligned} \tilde{V}_T^p(B_T^e, B_T^n, ps_T) = & \alpha_l^m \ln(l_T^m) + \alpha_l^f \ln(l_T^f) + \alpha_c \ln(c_T) - \xi(ps_T) + \\ & \beta^p \frac{\partial E_T V_{T+1}^p}{\partial(\ln k_{T+1})} (\ln R_T + \delta_T^m \ln \tau_T^m + \delta_T^f \ln \tau_T^f + \delta_T^c \ln \tau_T^c + \delta_T^e \ln e_T) \end{aligned}$$

where the expectation is taken over the final stock of child non-cognitive skills.

4. Loop over the K grid points for cognitive skills and add them back into the value function

$$V_T^p(B_T^e, B_T^n, ps_T) = \tilde{V}_T^p(B_T^e, B_T^n, ps_T) + \ln k_T (\alpha_k (1 - \varphi) + \beta^p \frac{\partial E_T V_{T+1}^p}{\partial(\ln k_{T+1})})$$

²⁹K is 35.

³⁰While this does not allow for dynamic complementarities in cognitive skills, [Ronda \(2017\)](#) uses a trans-log cognitive skill production function, and finds a statistically significant evidence of dynamic complementarity of mother's time with the child when the child is younger than 6 years old. This parameter is not statistically different from 0 when the child is between 6 and 16 years of age. Therefore, using similar data, [Ronda \(2017\)](#) does not provide significant evidence for the dynamic complementarity of parental time investment in the age range studied in this paper.

5. For each state (B_T^e, B_T^n, k_T) I compute the optimal choice, which yields $V_T^{p,*}(B_T^e, B_T^n, ps_T)$
6. To keep the backward induction process computationally tractable, I approximate $V_T^{p,*}(B_T^e, B_T^n, ps_T)$ using the following polynomial:

$$\hat{V}_T^p = z_{0,T} + z_{1,T} \ln k_T + z_{2,T} B_T^e + z_{3,T} B_T^n + z_{4,T} \ln k_T B_T^n + z_{5,T} \ln k_T B_T^e + z_{6,T} \ln k_T B_T^e B_T^n + z_{7,T} B_T^e B_T^n$$

Given OLS estimates for the z terms, the approximated value function is

$$\hat{V}_T^p = \hat{z}_{0,T} + \hat{z}_{1,T} \ln k_T + \hat{z}_{2,T} B_T^e + \hat{z}_{3,T} B_T^n + \hat{z}_{4,T} \ln k_T B_T^n + \hat{z}_{5,T} \ln k_T B_T^e + \hat{z}_{6,T} \ln k_T B_T^e B_T^n + \hat{z}_{7,T} B_T^e B_T^n$$

7. For any period $t \leq T$

$$\frac{\partial \hat{V}_t}{\partial \ln k_t} = \hat{z}_{1,t} + \hat{z}_{4,t} B_t^n + \hat{z}_{5,t} B_t^e + \hat{z}_{6,t} B_t^n B_t^e$$

8. Accounting for expectations:

$$\begin{aligned} E_{t-1} \frac{\partial \hat{V}_t}{\partial \ln k_t} &= \hat{z}_{1,t} + \hat{z}_{4,t} Pr(B_t^n = 1 | \Gamma_{t-1}, ps_{t-1}) + \\ &\quad \hat{z}_{5,t} Pr(B_t^e = 1 | \Gamma_{t-1}, ps_{t-1}) + \hat{z}_{6,t} Pr(B_t^n = 1 | \Gamma_{t-1}, ps_{t-1}) Pr(B_t^e = 1 | \Gamma_{t-1}, ps_{t-1}) \end{aligned}$$

This allows for the parents' marginal value of period t cognitive skills to be written as a function of known regression coefficients. In each step of the backward induction algorithm, I replace the parents' expected continuation value with the approximated one:

$$\begin{aligned} E_T(\hat{V}_t(B_t^n, B_t^e, k_t | a_t^p, \tau_t^c)) &= D_{t+1}^p \ln k_{t+1} + \hat{z}_{0,t} + \hat{z}_{4,t} E_t(B_{t+1}^e | \Gamma_t, a_t^p) + \\ &\quad + \hat{z}_{5,t} E_t(B_{t+1}^n | \Gamma_t, a_t^p) + \hat{z}_{7,t} E_t(B_{t+1}^e | \Gamma_t, a_t^p) E_t(B_{t+1}^n | \Gamma_t, a_t^p) \end{aligned}$$

9. With these regression coefficients in hand, I can iterate to period T-1, and the solution process remains the same

B Wages and Non-Labor Income

Table 15 shows the parameters governing the parent wage offer distribution. One can see that independent of age and education, father's have higher wage offers than mothers. Additionally, fathers see larger returns with respect to age, and slightly greater returns to education

as well.

Table 15. **Wage Offer Parameters**

Parameter	Estimate	(Std Error)
Father Intercept	1.375	(0.013)
Father Age Slope	0.012	(0.000)
Father Education	0.092	(0.001)
Mother Intercept	1.068	(0.012)
Mother Age Slope	0.009	(0.000)
Mother Education	0.090	(0.001)
Father Std Dev	0.572	(0.026)
Mother Std Dev	0.554	(0.029)
Correlation	0.799	(0.019)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

Table 17 shows that the model is able to generate the key patterns observed in the data with respect to parental labor supply and non-labor income, while Table 16 reports the parameters governing the non-labor income distribution. The probability of having any non-labor income is increasing at a decreasing rate with the age of the father, and also increases with the father's years of education. These patterns hold when looking at the level of non-labor income conditional on having a positive amount of non-labor income.

Table 16. **Non-Labor Income Parameters**

Parameter	Estimate	(Std Error)
Logit - Intercept	-7.258	(1.612)
Logit - Dad Age	0.248	(0.195)
Logit- Dad Age Squared	-0.003	(0.002)
Logit - Dad Educ	0.227	(0.312)
Conditional - Intercept	2.618	(0.528)
Conditional - Dad Age	0.043	(0.021)
Conditional - Dad Age Squared	-0.000	(0.000)
Conditional - Dad Educ	0.070	(0.042)
Conditional - SD of Shock	-0.056	(0.029)

Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

Table 17. **Wage and Non-Labor Income Fit**

	Data	Model
Avg Dad Wage	28.130	28.827
Avg Mom Wage	18.166	19.778
Avg Mom Hours	30.399	30.712
Avg Dad Hours	43.639	44.219
Frac Working Moms	0.835	0.884
Frac Working Dads	0.980	0.994
Avg NLI	150.049	155.054
SD NLI	205.424	213.528
Frac NLI > 0	0.828	0.835

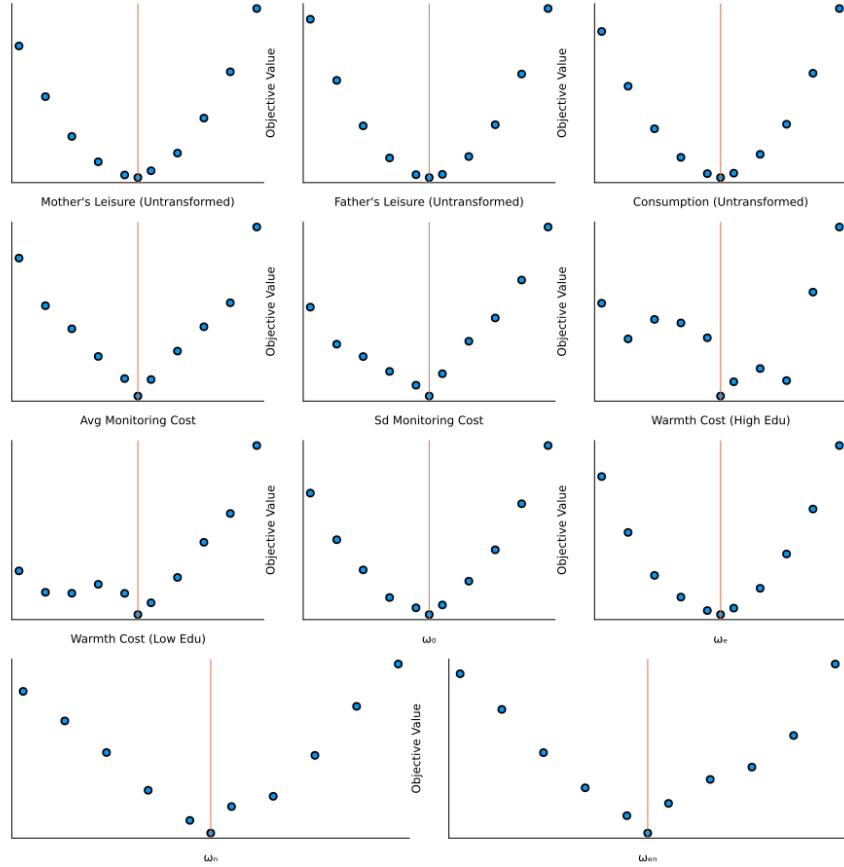
Notes: Standard errors are computed by bootstrapping households with replacement 50 times.

C Small Sample Discussion

While a sample size of 270 is slightly larger than similar works who use the PSID-CDS data, a discussion of potential sensitivity issues is still warranted. The most likely parameters to have sensitivity issues would be household preferences, as households face a common technology for the production of both types of non-cognitive skills, and cognitive skills are similar with the exception that parental education is allowed to influence the marginal product of time investment for parents and children. [Del Boca et al. \(2014\)](#) allowed for preference heterogeneity in a sample of 105 households, which could be sensitive to small changes in the parameters governing the standard deviation of these parameters. To alleviate this issue, I assume that all parents have common preferences over leisure, consumption, and their child's stock of non-cognitive skills. Children also have four common potential outcomes for their leisure preferences, but these are allowed to change over time as their non-cognitive skills evolve.

In Figure 4 below, I plot the value of the SMM objective function after individually perturbing the value of the preference parameters from 10% smaller to 10% larger than the structural parameter estimate, which is denoted by the vertical line within each plot. For the parameters governing parent leisure and consumption (and therefore child skills), there is a smooth trend away as I perturb away from the parameter estimate, so these seem to be identified. This also holds for the average and standard deviation of the parental monitoring costs, as well as the child's preference parameters. The only two parameters where I do not observe this smooth trend are with respect to the parental warmth costs for high and low educated mothers, however, the parameter estimate is still the lowest point in the region, and the objective function is trending upward as I move away from the parameter estimate.

Figure 4. **Sensitivity Analysis**

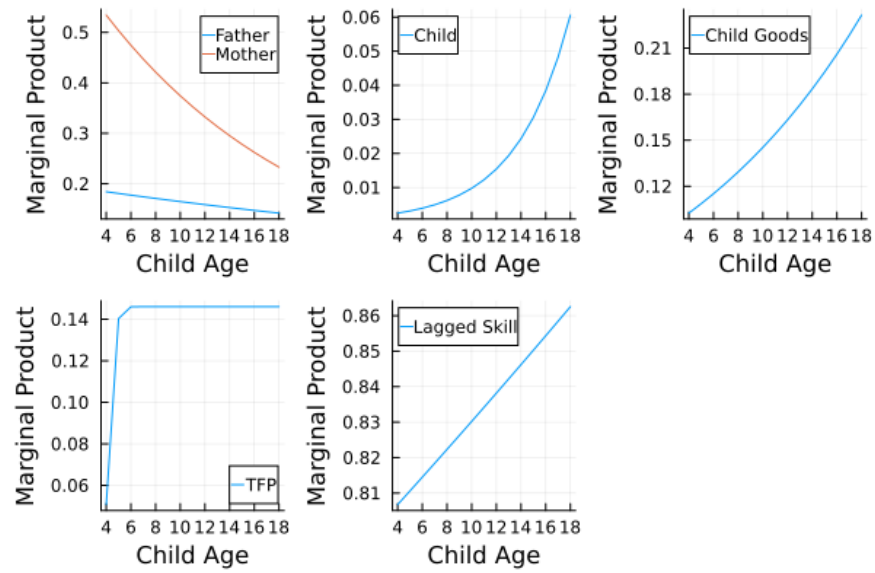


Notes: This figure is produced by perturbing the structural parameter estimates by plus or minus ten percent of, simulating the model, and plotting the value of the SMM objective function.

D Productivity Plots

Figure 5 displays the marginal products for the components of the child cognitive skill production function (where education has been removed for clarity). Consistent with the literature, the marginal product of maternal time is higher than the marginal product for paternal time, and these values decrease with the age of children. The marginal product of child time investment increases with age, and this is true for goods investment as well as lagged cognitive skills. TFP levels off very early, around when the child is six years of age.

Figure 5. **Cognitive Skill Productivity**



Notes: This figure is produced by perturbing the structural parameter estimates by plus or minus ten percent of, simulating the model, and plotting the value of the SMM objective function.

E Moment Conditions

Time Allocation

	Moment	Data	Model
Mother's Avg Time Investment age 4-5		25.78	23.67
Mother's Avg Time Investment age 6-8		18.83	20.56
Mother's Avg Time Investment age 9-11		16.12	18.64
Mother's Avg Time Investment age 12-15		14.32	16.00
Mother's Avg Time Investment age 16-18		12.79	12.81
Mother's SD Time Investment age 4-5		13.22	4.79
Mother's SD Time Investment age 6-8		11.03	5.13
Mother's SD Time Investment age 9-11		12.44	4.84
Mother's SD Time Investment age 12-15		12.12	4.21
Mother's SD Time Investment age 16-18		12.06	2.84
Father's Avg Time Investment age 4-5		10.39	8.15
Father's Avg Time Investment age 6-8		8.34	7.90
Father's Avg Time Investment age 9-11		8.13	7.90
Father's Avg Time Investment age 12-15		6.93	7.77
Father's Avg Time Investment age 16-18		5.80	7.21
Father's SD Time Investment age 4-5		9.43	1.30
Father's SD Time Investment age 6-8		9.11	1.70
Father's SD Time Investment age 9-11		10.62	1.67
Father's SD Time Investment age 12-15		9.70	1.60
Father's SD Time Investment age 16-18		12.95	1.29
Child's Avg Time Investment age 4-5		0.61	0.68
Child's Avg Time Investment age 6-8		1.32	1.53
Child's Avg Time Investment age 9-11		3.57	2.91
Child's Avg Time Investment age 12-15		6.22	5.02
Child's Avg Time Investment age 16-18		5.83	6.40
Child's SD Time Investment age 4-5		1.17	1.05
Child's SD Time Investment age 6-8		2.31	1.89
Child's SD Time Investment age 9-11		4.28	3.35
Child's SD Time Investment age 12-15		7.65	4.97
Child's SD Time Investment age 16-18		7.80	5.66
Avg Study Time Authoritarian Age 4-11		2.25	1.93
Avg Study Time Authoritarian Age 12-18		5.86	5.25
Avg Study Time Authoritative Age 4-11		2.44	2.18
Avg Study Time Authoritative Age 12-18		6.24	5.92
Avg Study Time Permissive Age 4-11		2.36	2.55
Avg Study Time Permissive Age 12-18		6.04	5.59
Avg Study Time Authoritarian		4.37	4.32
Avg Study Time Authoritative		4.37	4.20
Avg Study Time Permissive		4.48	4.47

	Moment	Data	Model
Father's Avg Hours Worked age 4-5		43.09	42.88
Father's Avg Hours Worked age 6-8		43.78	42.69
Father's Avg Hours Worked age 9-11		43.79	43.58
Father's Avg Hours Worked age 12-15		43.92	44.16
Father's Avg Hours Worked age 16-18		43.52	45.77
Father's SD Hours Worked age 4-5		10.48	10.32
Father's SD Hours Worked age 6-8		9.17	10.27
Father's SD Hours Worked age 9-11		8.98	10.35
Father's SD Hours Worked age 12-15		9.81	10.48
Father's SD Hours Worked age 16-18		10.58	10.42
Mother's Avg Hours Worked age 4-5		28.81	25.96
Mother's Avg Hours Worked age 6-8		29.11	27.94
Mother's Avg Hours Worked age 9-11		29.88	28.77
Mother's Avg Hours Worked age 12-15		31.77	31.26
Mother's Avg Hours Worked age 16-18		32.73	33.41
Mother's SD Hours Worked age 4-5		13.50	13.28
Mother's SD Hours Worked age 6-8		13.40	13.32
Mother's SD Hours Worked age 9-11		12.60	13.72
Mother's SD Hours Worked age 12-15		12.72	14.14
Mother's SD Hours Worked age 16-18		12.48	14.57
Fraction Father's Working Age 4-5		0.99	0.99
Fraction Father's Working Age 6-8		0.98	0.99
Fraction Father's Working Age 9-11		0.98	0.99
Fraction Father's Working Age 12-15		0.98	0.99
Fraction Father's Working Age 16-18		0.95	1.00
Fraction Mother's Working Age 4-5		0.80	0.81
Fraction Mother's Working Age 6-8		0.81	0.86
Fraction Mother's Working Age 9-11		0.85	0.87
Fraction Mother's Working Age 12-15		0.86	0.89
Fraction Mother's Working Age 16-18		0.87	0.92
Correlation of Education and Mother's Time age 4-5		0.10	0.10
Correlation of Education and Mother's Time age 6-8		-0.04	0.08
Correlation of Education and Mother's Time age 9-11		-0.09	0.08
Correlation of Education and Mother's Time age 12-15		0.07	0.06
Correlation of Education and Mother's Time age 16-18		0.00	0.06
Correlation of Education and Father's Time age 4-5		0.06	0.07
Correlation of Education and Father's Time age 6-8		0.11	0.05
Correlation of Education and Father's Time age 9-11		0.05	0.01
Correlation of Education and Father's Time age 12-15		-0.06	-0.03
Correlation of Education and Father's Time age 16-18		-0.12	-0.04

	Moment	Data	Model
Correlation of Mother's Education and Child's Time age 4-5		0.10	0.07
Correlation of Mother's Education and Child's Time age 6-8		0.03	0.11
Correlation of Mother's Education and Child's Time age 9-11		0.18	0.09
Correlation of Mother's Education and Child's Time age 12-15		0.17	0.07
Correlation of Mother's Education and Child's Time age 16-18		0.39	0.08
Avg Child Study Time BPIE = 1		4.28	5.47
Avg Child Study Time BPIE = 0		3.94	1.88
Avg Child Study Time BPIN = 1		3.86	2.69
Avg Child Study Time BPIN = 0		4.67	6.40
Avg Child Study Time BPIE = 1, Permissive Style		4.92	5.51
Avg Child Study Time BPIE = 0, Permissive Style		3.64	0.80
Avg Child Study Time BPIN = 1, Permissive Style		4.14	2.43
Avg Child Study Time BPIN = 0, Permissive Style		4.91	7.03
Avg Child Study Time BPIN = 0 BPIE = 0		4.51	1.69
Avg Child Study Time BPIN = 0 BPIE = 1		4.93	12.62
Avg Child Study Time BPIN = 1 BPIE = 0		2.69	2.51
Avg Child Study Time BPIN = 1 BPIE = 1		4.10	2.71
Avg Mother Time Investment Authoritarian		14.53	15.48
Avg Mother Time Investment Authoritative		16.71	17.16
Avg Mother Time Investment Permissive		17.51	16.83
Avg Father Time Investment Authoritarian		7.24	7.39
Avg Father Time Investment Authoritative		7.53	7.74
Avg Father Time Investment Permissive		7.11	7.87
SD Child Time Investment Authoritarian		6.55	4.36
SD Child Time Investment Authoritative		6.37	4.72
Avg Child Time Conditional Intensive, HS or Less		3.62	4.24
Avg Child Time Conditional Intensive, Some College		4.26	4.29

Parenting Style

	Moment	Data	Model
Fraction Authoritarian		0.27	0.27
Fraction Authoritative		0.46	0.45
Fraction Permissive		0.27	0.28
Fraction Authoritarian More Than HS		0.25	0.25
Fraction Authoritative More Than HS		0.47	0.47
Fraction Permissive More Than HS		0.28	0.28
Fraction Authoritarian HS		0.30	0.31
Fraction Authoritative HS		0.45	0.42
Fraction Permissive HS		0.25	0.27
Fraction Authoritarian Age 4-10		0.21	0.19
Fraction Authoritative Age 4-10		0.50	0.54
Fraction Permissive Age 4-10		0.28	0.27
Fraction Authoritarian Age 11-18		0.30	0.30
Fraction Authoritative Age 11-18		0.43	0.42
Fraction Permissive Age 11-18		0.27	0.28
Fraction Always Intensive		0.77	0.93
Fraction Always Warm		0.57	0.72
Fraction Always Cold		0.53	0.63

Cognitive Skills

	Moment	Data	Model
	Avg LW Score Age 4	9.31	9.31
	Avg LW Score Age 5	12.62	11.77
	Avg LW Score Age 6	22.77	22.61
	Avg LW Score Age 7	32.32	31.66
	Avg LW Score Age 8	38.03	37.98
	Avg LW Score Age 9	42.32	42.20
	Avg LW Score Age 10	44.33	44.79
	Avg LW Score Age 11	44.25	46.41
	Avg LW Score Age 12	48.46	47.84
	Avg LW Score Age 13	47.76	48.77
	Avg LW Score Age 14	47.91	49.55
	Avg LW Score Age 15	50.17	50.30
	Avg LW Score Age 16	50.55	51.07
	Avg LW Score Age 17	51.39	51.82
	Avg LW Score Age 18	51.95	52.76
	SD LW Score Age 4-8	12.94	12.45
	SD LW Score Age 9-11	5.43	5.97
	SD LW Score Age 12-18	4.18	4.46
	Correlation of Child Time Investment and LW Score	0.33	0.37
	Correlation of Mother's Investment and LW Score	-0.29	-0.36
	Correlation of Father's Investment and LW Score	-0.09	-0.00
	Correlation of LW score in t with LW score in t-5	-0.96	-0.94
	Correlation of Father's Wage in 1998 and LW Score in 2002	0.16	0.19
	Correlation of Father's Wage in 1998 and LW Score in 2002	0.10	0.14
	Correlation of Father's Wage in 1998 and Change in LW Score	0.06	-0.02
	Correlation of Mother's Wage in 1998 and Change in LW Score	0.06	-0.00
	Correlation of Mother's Time With Change in LW Score	0.29	0.42
	Correlation of Father's Time With Change in LW Score	0.09	0.03
	Correlation of Child's Time With Change in LW Score	-0.33	-0.33
	Correlation of Mother's Time With Change in LW Score Age 4-7	0.18	0.57
	Correlation of Mother's Time With Change in LW Score Age 8-11	0.09	-0.06
	Correlation of Father's Time With Change in LW Score Age 4-7	0.20	0.44
	Correlation of Father's Time With Change in LW Score Age 8-11	-0.07	-0.11
	Correlation of Child's Time With Change in LW Score Age 4-7	-0.21	-0.38
	Correlation of Child's Time With Change in LW Score Age 8-11	-0.20	-0.15
	Correlation of Lagged LW Score With Change in LW Score Age 4-7	-0.90	-0.85
	Correlation of Lagged LW Score With Change in LW Score Age 8-11	-0.74	-0.74
	Correlation of Father's Wage in 1998 and Change in LW Score Age 4-7	0.01	0.11
	Correlation of Father's Wage in 1998 and Change in LW Score Age 8-11	-0.03	0.00
	Correlation of Mother's Wage in 1998 and Change in LW Score Age 4-7	-0.00	0.06
	Correlation of Mother's Wage in 1998 and Change in LW Score Age 8-11	-0.05	0.02

Wages

	Moment	Data	Model
Avg Father's Wage if Working		28.13	28.83
St Dev Father's Wage if Working		21.46	19.78
Avg Mother's Wage if Working		18.17	18.44
St Dev Mother's Wage if Working		12.87	11.83
Avg Father's Wage if Working Age 0-32		18.87	20.96
Avg Father's Wage if Working Age 33-37		23.29	23.32
Avg Father's Wage if Working Age 38-42		27.01	25.95
Avg Father's Wage if Working Age 43-75		32.68	30.94
Avg Mother's Wage if Working Age 0-32		12.15	14.14
Avg Mother's Wage if Working Age 33-37		15.77	15.63
Avg Mother's Wage if Working Age 38-42		18.60	17.46
Avg Mother's Wage if Working Age 43-75		21.00	19.90
St Dev Father's Wage if Working Age 0-32		16.10	14.18
St Dev Father's Wage if Working Age 33-37		13.91	13.57
St Dev Father's Wage if Working Age 38-42		19.01	17.20
St Dev Father's Wage if Working Age 43-75		25.18	21.15
St Dev Mother's Wage if Working Age 0-32		6.91	8.49
St Dev Mother's Wage if Working Age 33-37		11.30	9.68
St Dev Mother's Wage if Working Age 38-42		12.85	10.89
St Dev Mother's Wage if Working Age 43-75		14.18	12.67
Avg Father's Wage if Working Education 0-12		20.51	22.59
Avg Father's Wage if Working Education 13-15		25.78	27.58
Avg Father's Wage if Working Education 16-50		39.31	36.92
Avg Mother's Wage if Working Education 0-12		13.38	14.48
Avg Mother's Wage if Working Education 13-15		17.96	17.90
Avg Mother's Wage if Working Education 16-50		23.83	22.96
Avg Father's Wage if Less Than HS and Younger Than 40		19.49	20.72
Avg Father's Wage if Less Than HS and Older Than 40		21.42	23.21
Avg Mother's Wage if Less Than HS and Younger Than 40		12.38	13.33
Avg Mother's Wage if Less Than HS and Older Than 40		14.67	15.08
Avg Father's Wage if Less Than College and Younger Than 40		21.24	24.75
Avg Father's Wage if Less Than College and Older Than 40		29.20	28.58
Avg Mother's Wage if Less Than College and Younger Than 40		16.58	16.73
Avg Mother's Wage if Less Than College and Older Than 40		19.43	18.49
Avg Father's Wage if College and Younger Than 40		33.16	30.62
Avg Father's Wage if College and Older Than 40		41.62	37.65
Avg Mother's Wage if College and Younger Than 40		21.14	20.48
Avg Mother's Wage if College and Older Than 40		24.87	23.32
Correlation of Accepted Wages		0.22	0.83
Pct of Income Invested in Child		21.00	20.44

Non-Cognitive Skills

	Moment	Data	Model
	Avg BPIE	0.66	0.68
	Avg BPIN	0.62	0.56
	Avg BPIE if Lagged BPIE = 1	0.85	0.81
	Avg BPIE if Lagged BPIE = 0	0.42	0.68
	Avg BPIE if Lagged BPIN = 1	0.78	0.86
	Avg BPIE if Lagged BPIN = 0	0.55	0.68
	Avg BPIE if Lagged BPIE = 1, Lagged Style Authoritarian	0.84	0.80
	Avg BPIE if Lagged BPIE = 1, Lagged Style Authoritative	0.83	0.81
	Avg BPIE if Lagged BPIE = 1, Lagged Style Permissive	0.88	0.83
	Avg BPIE if Lagged BPIE = 0, Lagged Style Authoritarian	0.44	0.59
	Avg BPIE if Lagged BPIE = 0, Lagged Style Authoritative	0.44	0.62
	Avg BPIE if Lagged BPIE = 0, Lagged Style Permissive	0.38	0.88
	Avg BPIE if Lagged BPIN = 1, Lagged Style Authoritarian	0.73	0.71
	Avg BPIE if Lagged BPIN = 1, Lagged Style Authoritative	0.78	0.87
	Avg BPIE if Lagged BPIN = 1, Lagged Style Permissive	0.83	0.99
	Avg BPIE if Lagged BPIN = 0, Lagged Style Authoritarian	0.55	0.61
	Avg BPIE if Lagged BPIN = 0, Lagged Style Authoritative	0.59	0.62
	Avg BPIE if Lagged BPIN = 0, Lagged Style Permissive	0.50	0.83
	Avg BPIN if Lagged BPIN = 1	0.75	0.90
	Avg BPIN if Lagged BPIN = 0	0.40	0.53
	Avg BPIN if Lagged BPIE = 1	0.72	0.68
	Avg BPIN if Lagged BPIE = 0	0.42	0.59
	Avg BPIN if Lagged BPIN = 1, Lagged Style Authoritarian	0.79	0.82
	Avg BPIN if Lagged BPIN = 1, Lagged Style Authoritative	0.70	0.91
	Avg BPIN if Lagged BPIN = 1, Lagged Style Permissive	0.78	0.95
	Avg BPIN if Lagged BPIN = 0, Lagged Style Authoritarian	0.45	0.50
	Avg BPIN if Lagged BPIN = 0, Lagged Style Authoritative	0.41	0.54
	Avg BPIN if Lagged BPIN = 0, Lagged Style Permissive	0.33	0.52
	Avg BPIN if Lagged BPIE = 1, Lagged Style Authoritarian	0.81	0.71
	Avg BPIN if Lagged BPIE = 1, Lagged Style Authoritative	0.68	0.70
	Avg BPIN if Lagged BPIE = 1, Lagged Style Permissive	0.70	0.62
	Avg BPIN if Lagged BPIE = 0, Lagged Style Authoritarian	0.47	0.57
	Avg BPIN if Lagged BPIE = 0, Lagged Style Authoritative	0.40	0.59
	Avg BPIN if Lagged BPIE = 0, Lagged Style Permissive	0.41	0.60
	Fraction Authoritarian if BPIE = 1	0.26	0.23
	Fraction Authoritative if BPIE = 1	0.48	0.45
	Fraction Permissive if BPIE = 1	0.27	0.32
	Fraction Authoritarian if BPIE = 0	0.31	0.36
	Fraction Authoritative if BPIE = 0	0.42	0.46
	Fraction Permissive if BPIE = 0	0.27	0.19

	Moment	Data	Model
Fraction Authoritarian if BPIN = 1		0.30	0.31
Fraction Authoritative if BPIN = 1		0.45	0.42
Fraction Permissive if BPIN = 1		0.25	0.27
Fraction Authoritarian if BPIN = 0		0.23	0.22
Fraction Authoritative if BPIN = 0		0.46	0.50
Fraction Permissive if BPIN = 0		0.31	0.28
Avg BPIE if Age Less than 12		0.63	0.64
Avg BPIN if Age Less than 12		0.61	0.50
Avg BPIE Age 12 and up		0.69	0.70
Avg BPIN Age 12 and Up		0.63	0.60